

ETW3420 Principles of Forecasting and Applications

Group Assignment 2: Forecasting Energy Consumption Patterns Analysis of U.S. Transportation Energy Consumption During and After COVID-19

Group: Group 3

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Stage 1: Data Collection and Preparation

```
# Load dataset
ts_data <- read.csv("MER_T02_01B.csv")

# Remove month 13 (annual totals)
df <- subset(ts_data, as.integer(substr(as.character(YYYYMM), 5, 6)) != 13)

# Remove missing values
df <- na.omit(df)

# Parse Year and Month
df$Year <- as.integer(substr(as.character(df$YYYYMM), 1, 4))
df$Month <- as.integer(substr(as.character(df$YYYYMM), 5, 6))

# Filter to 2010–2025 only
df <- subset(df, Year >= 2010 & Year <= 2025)

# Extract dataset for a given MSN code and convert directly to time series
make_ts <- function(data, msn_code) {
  out <- subset(data, MSN == msn_code)
  out <- out[order(out$Year, out$Month), ]
  ts(out$value,
     start = c(out$Year[1], out$Month[1]),
     frequency = 12)
}

ts_TotalPrimaryEnergy_Trans <- make_ts(df, "TXACBUS")
ts_ElecSales <- make_ts(df, "ESACBUS")
ts_EndUseEnergy <- make_ts(df, "TNACBUS")
ts_ElecSystemLoss <- make_ts(df, "LOACBUS")
ts_TotalEnergy <- make_ts(df, "TEACBUS")
ts_TotalPrimaryEnergy_End <- make_ts(df, "TETXBUS")
```

→ `na.omit()`: Removes missing values to ensure clean data

→ `make_ts()`: Creates monthly times series objects for each dataset

→ `ts()`: Converts the data into time series format with monthly frequency

In this stage, the dataset from the U.S. Energy Information Administration (EIA) was imported and prepared for time series analysis. Data cleaning involved removing observations with month = 13 (annual totals) to avoid distorting seasonal patterns and addressing missing values. The dataset was then filtered to the period 2010–2015 to ensure sufficient observations for modelling while excluding older structural breaks such as the 1970s oil crisis and the 2008 financial crisis. Importantly, this interval also captures the COVID-19 shock in 2020, a key structural break in energy consumption, particularly in the transport sector. This ensures the dataset is clean, consistent, and suitable for forecasting analysis under both normal and shock conditions.

Stage 2: Exploration and Visualization

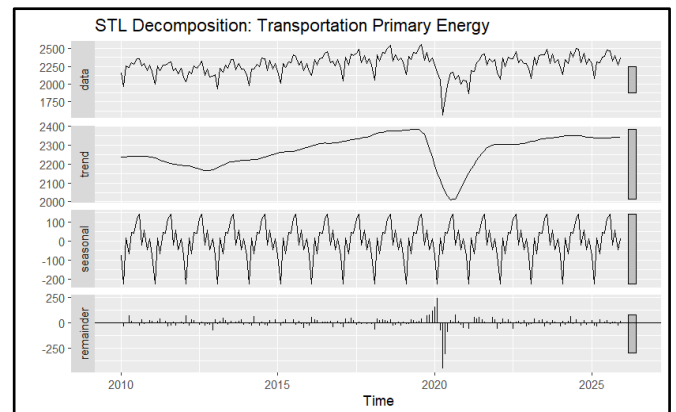
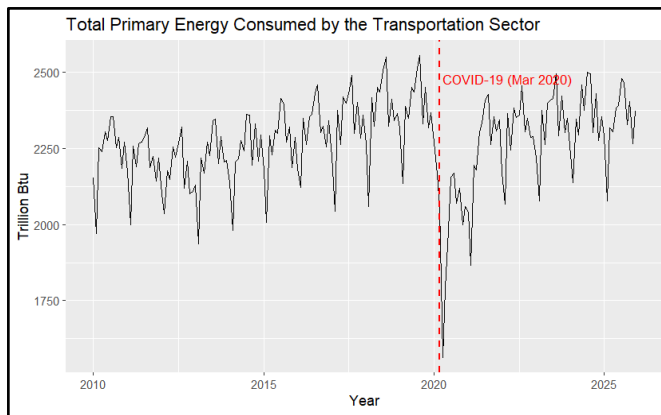
DATASET 1: Total Primary Energy Consumed by the Transportation Sector (TXACBUS)

2.1 Time Series Plot & STL Decomposition

```
# Time series plot
autoplot(ts_TotalPrimaryEnergy_Trans) +
  ggtitle("Total Primary Energy Consumed by the Transportation Sector") +
  xlab("Year") +
  ylab("Trillion Btu") +
  geom_vline(xintercept = 2020 + 2/12, colour = "red",
             linetype = "dashed", size = 0.8) +
  annotate("text", x = 2020 + 3/12,
          y = max(ts_TotalPrimaryEnergy_Trans) * 0.97,
          label = "COVID-19 (Mar 2020)",
          colour = "red", hjust = 0, size = 3.5)

# Decomposition (STL)
ts_TotalPrimaryEnergy_Trans %>%
  stl(s.window = "periodic") %>%
  autoplot() +
  ggtitle("STL Decomposition: Transportation Primary Energy")
```

→ `autoplot()`: plot the full series with COVID-19 marker in March 2020

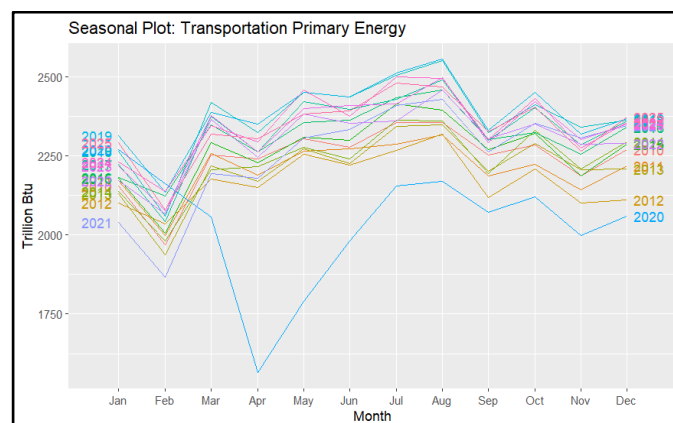


The transportation primary energy series displays a strong seasonal pattern (period = 12) and a clear upward trend from 2010 to 2019. A major structural break occurred in March 2020, when COVID-19 lockdowns caused consumption to plunge approximately 37% to below 1,650 trillion Btu, with the EIA (2021) reporting a record 15% annual decline driven by grounded aviation and suppressed road traffic. Recovery through 2021–2022 was swift but incomplete, with post-pandemic levels stabilizing around 2,250–2,400 Trillion Btu, reflecting lasting behavioural shifts such as remote working and reduced business travel. STL decomposition confirms that COVID-19 disrupted overall consumption levels rather than the underlying seasonal structure.

2.2 Seasonal Plot

```
# Seasonal plot
ggseasonplot(ts_TotalPrimaryEnergy_Trans, year.labels = TRUE, year.labels.left = TRUE) +
  ggtitle("Seasonal Plot: Transportation Primary Energy") +
  xlab("Month") +
  ylab("Trillion Btu")
```

→ **ggseasonplot():**
Compares monthly
seasonal patterns across

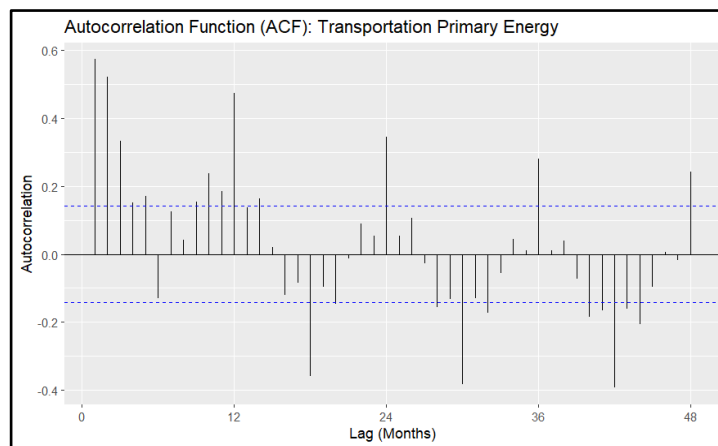


Pre-COVID years (2012–2019) cluster tightly between 2,050–2,500 Trillion Btu with a gradual upward drift, with energy use peaking in July–August and troughing in November–December, consistent with the US summer driving season where gasoline and jet fuel demand peaks from April to September (EIA, 2020). The 2020 line is a clear outlier, collapsing to approximately 1,650 trillion Btu in April before gradually recovering to the pre-pandemic cluster by 2022–2025.

2.3 ACF Plot

```
# ACF
ggAcf(ts_TotalPrimaryEnergy_Trans, lag.max = 48) +
  ggtitle("Autocorrelation Function (ACF): Transportation Primary Energy") +
  xlab("Lag (Months)") +
  ylab("Autocorrelation")]
```

→ **ggAcf():** Plots autocorrelations at different lags



A strong positive autocorrelation at short lags with slow decay indicate a persistent trend and non-stationarity, suggesting the need for first differencing ($d = 1$), while repeating spikes at lags 12 and 24 confirm strong annual seasonality. Thus, this supports seasonal differencing ($d = 1$, seasonal period = 12) for ARIMA modelling.

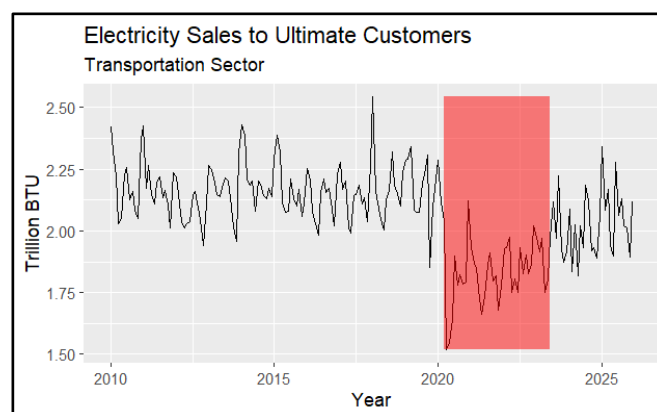
DATASET 2: Electricity Sales to Ultimate Customers in the Transportation Sector (ESACBUS)

2.1 Time Series Plot

```
{r}
# converting the dataframe type to timeseries
ts_ElecSales <- ts(dataset2$Value, start = c(dataset2$Year[1], dataset2$Month[1]), frequency = 12)

autoplot(ts_ElecSales) +
  annotate("rect", xmin = 2020 + 2/12, xmax = 2023 + 5/12,
         ymin = min(ts_ElecSales), ymax = max(ts_ElecSales),
         fill = "red", alpha = 0.5) +
  ggtitle("Electricity Sales to Ultimate Customers") +
  labs(subtitle = "Transportation Sector", x = "Year", y = "Trillion BTU")
```

→ **time series plot:** Visualise the overall pattern and behaviour of the data over time, enabling clear identification of trends and fluctuations. The COVID-19 period is

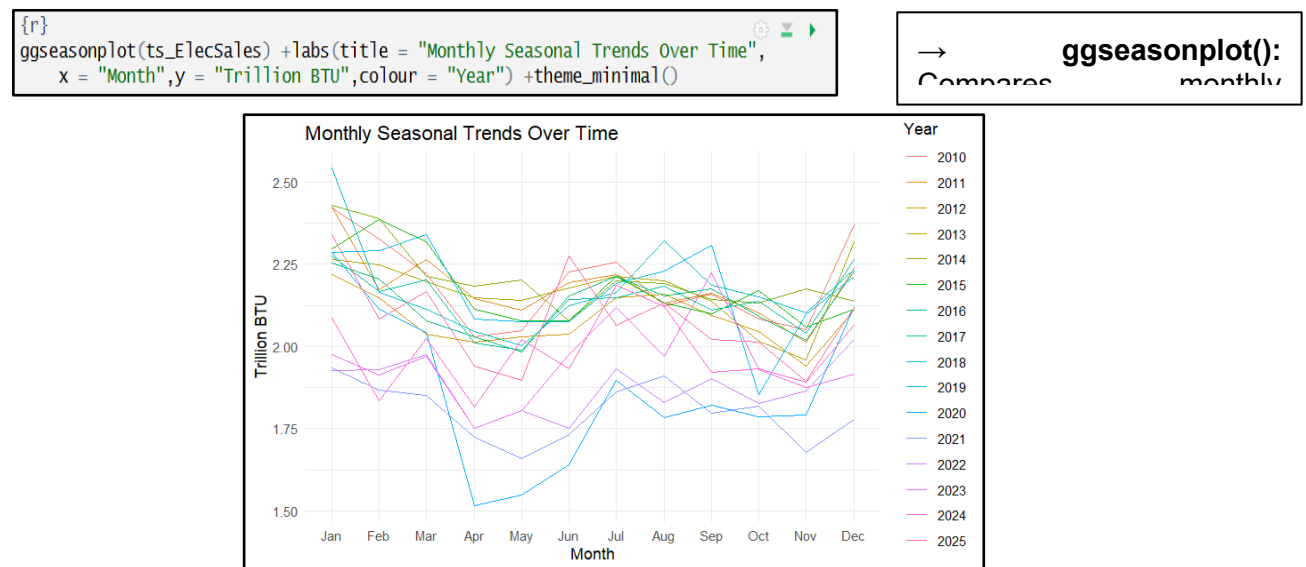


From 2010 up to the onset of COVID-19, the series remained relatively stable with no strong long-term growth pattern before the pandemic. A sharp decline in early 2020, reflecting a

strong impact on the electricity demand in the transportation sector. However, the series remains low during the COVID period and does not return to pre-pandemic level. From around mid-2023 onwards, a gradual recovery is observed, though levels remain generally lower than before the pandemic.

Thus, there is no clear long-term trend or structural cycle, but the series exhibits persistent short-term seasonal fluctuations that remain consistent throughout, suggesting stable seasonal drivers such as weather pattern or travel behaviour even amid major disruption.

2.2 Seasonal Plot



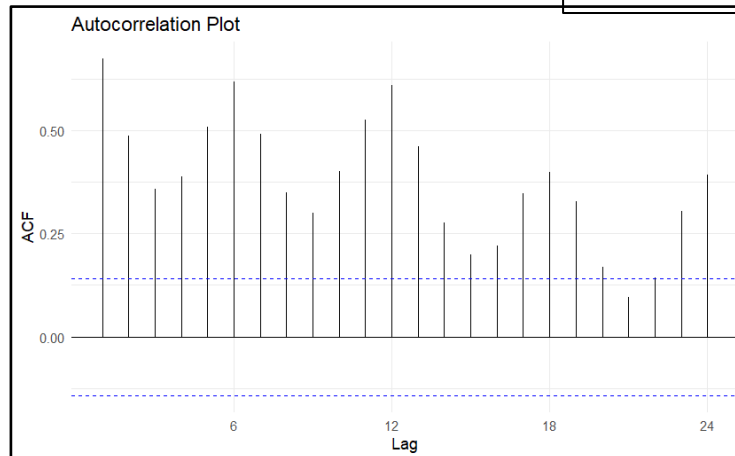
The plot reveals a bi-modal seasonal pattern with primary peaks in December and January and a secondary hump in July. These increases are associated with peak holiday travel and higher energy needs for heating or cooling in electric transit. In contrast, demand consistently declines during the shoulder months of April and November, when milder weather reduces energy use and travel activity is relatively subdued between major holiday periods.

Notably, the 2020–2022 period shows a sharp structural decline due to the pandemic. While 2024 and 2025 show a recovery from those lows, demand has not yet returned to the pre-pandemic levels, suggesting a persistent reduction in overall industry activity.

2.3 ACF Plot

```
{r}
ggAcf(ts_ElecSales)+labs(title = "Autocorrelation Plot") +theme_minimal()
```

→ **ggAcf()**: shows the correlation of the current data with its past



The ACF plot reveals a clear yearly seasonal pattern with distinct spikes repeating every 12 months, confirming that demand follows a stable and predictable seasonal cycle where past-year values strongly influence current observations. Although there is no consistent upward trend, the slow decay of the autocorrelations indicates strong persistence and non-stationarity. This suggests that the series is highly dependent on its past values and has been affected structurally by the level shift during the COVID-19 pandemic.

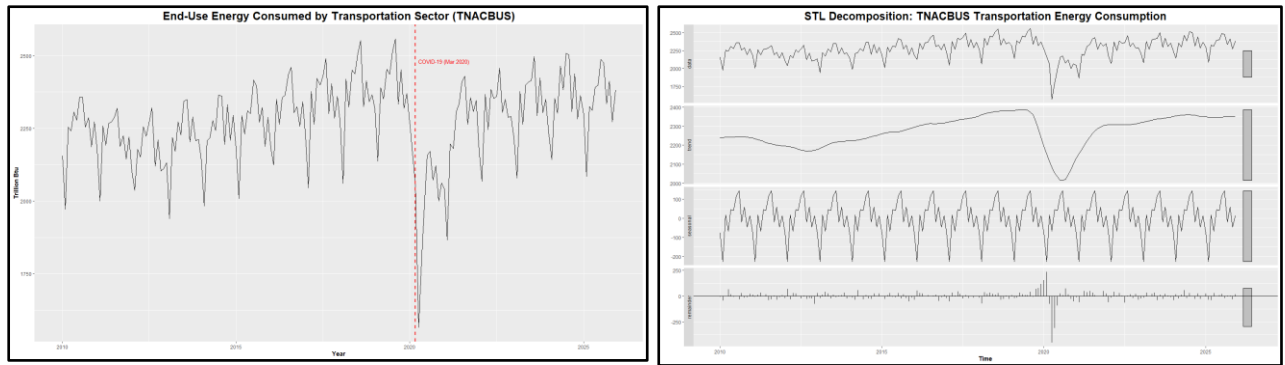
DATASET 3 : End-use energy consumption in the transportation sector (TNACBUS)

2.1 Time Series Plot

```
# STAGE 2: Visualisation ####
## 1. Time Series Plot (Overall Trend)###
autoplot(TNACBUS_ts) +
  ggtitle("End-use Energy Consumed by Transportation Sector (TNACBUS)") +
  xlab("Year") +
  ylab("Trillion Btu") +
  geom_vline(xintercept = 2020 + 2/12,
    colour = "red",
    linetype = "dashed",
    size = 0.8) +
  annotate("text",
    x = 2020 + 3/12,
    y = max(TNACBUS_ts, na.rm = TRUE) * 0.97,
    label = "COVID-19 (Mar 2020)",
    colour = "red",
    hjust = 0,
    size = 3.5) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 20),
    axis.title = element_text(face = "bold")
  )

## 2. STL Decomposition ####
TNACBUS_ts %>%
  stl(s.window = "periodic") %>%
  autoplot() +
  ggtitle("STL Decomposition: TNACBUS Transportation Energy Consumption") +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 20),
    axis.title = element_text(face = "bold")
  )
```

→ **autoplot()**: displays the time series of transportation energy consumption with a vertical line and annotation marking the COVID-19 period for clearer interpretation of structural changes



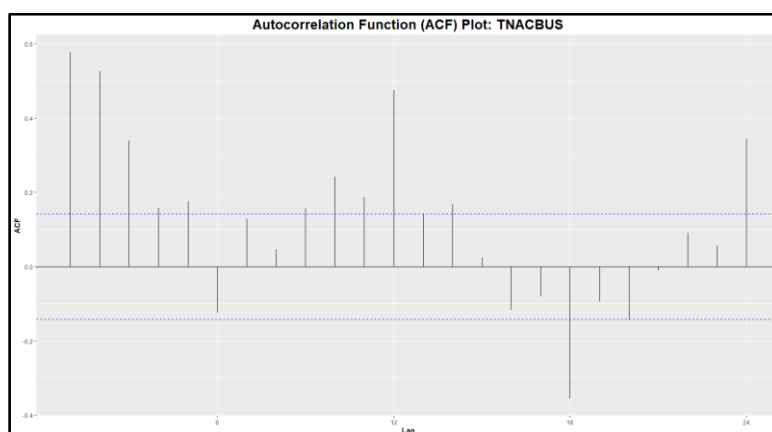
The time series plot of TNACBUS shows a moderate upward movement with cyclical fluctuations in transportation energy consumption prior to 2020 followed by a sharp structural break during the COVID-19 period due to lockdowns and reduced mobility. After 2020, the series gradually recovers but remains below pre-pandemic levels, suggesting possible long-term behavioural changes in transportation demand. Thus, the plot suggests the presence of trend, seasonality and structural changes, indicating that the series is non-stationary.

Next, the STL decomposition further confirms these characteristics, showing a clear upward trend before a pronounced drop during COVID-19 and a gradual post-pandemic recovery. Also, the seasonal component remains stable and consistent, indicating strong and persistent annual patterns, while the residuals capture heightened irregular shocks around 2020. Thus, the decomposition highlights a strong trend, stable seasonality and significant structural break in the series.

2.3 ACF Plot

```
## 2. ACF Plot ###
ggAcf(TNACBUS_ts) +
  ggtitle("Autocorrelation Function (ACF) Plot: TNACBUS") +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 20),
    axis.title = element_text(face = "bold")
  )
```

→ **ggACF()**: measures and visualises the correlation between the current value of a time series and



The ACF plot shows strong positive autocorrelation at short lags, indicating that current transportation energy consumption is strongly influenced by recent past values. The significant

spikes at lag 12 confirms a clear annual seasonal pattern, where observations are strongly correlated with the same month in previous years. In addition, the slow decay of the autocorrelation function suggests the presence of non-stationarity driven by underlying trend and seasonal effects. Thus, these patterns indicate that the series contains both trend and seasonal structures, requiring both first and seasonal differencing before fitting an ARIMA model.

DATASET 4: Electrical System Energy Losses Proportioned to the Transportation Sector (LOACBUS)

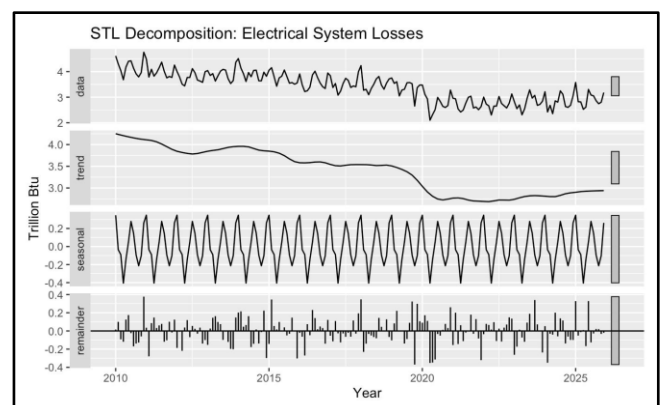
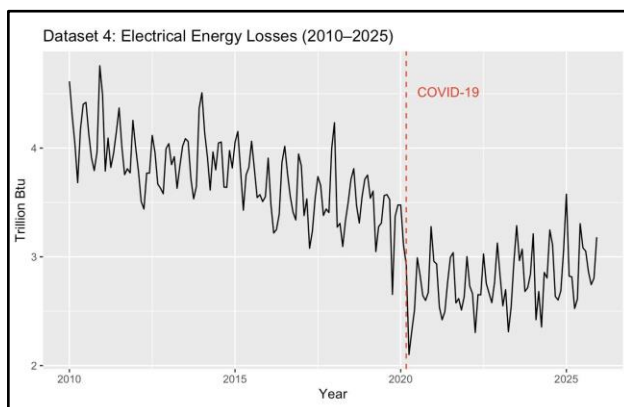
2.1 Time Series Plot & STL Decomposition

```
autoplot(ts_ElecSystemLoss) +
  ggtitle("Dataset 4: Electrical Energy Losses (2010–2025)") +
  xlab("Year") +
  ylab("Trillion Btu") +
  geom_vline(xintercept = 2020 + 2/12,
    colour = "red",
    linetype = "dashed") +
  annotate("text",
    x = 2020.5,
    y = max(ts_ElecSystemLoss) * 0.95,
    label = "COVID-19",
    colour = "red",
    hjust = 0)

# 2.2 STL Decomposition
ts_ElecSystemLoss %>%
  stl(s.window = "periodic") %>%
  autoplot() +
  ggtitle("STL Decomposition: Electrical System Losses") +
  xlab("Year") +
  ylab("Trillion Btu")
```

→ **autoplot ()**: used to visualise the overall behaviour of the series

→ **stl()**: applied to decompose the data into trend, seasonal and irregular components for



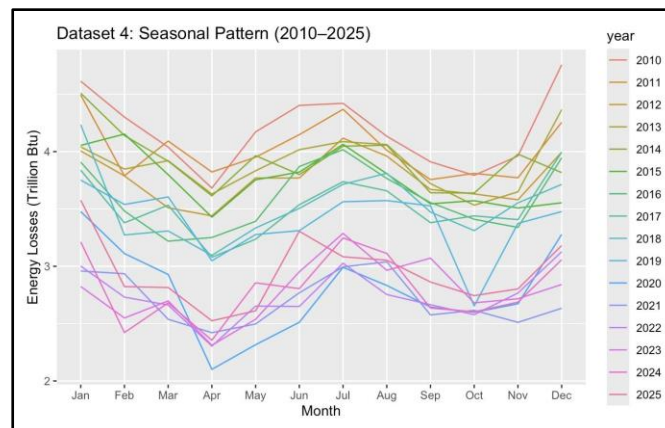
The time series plot shows clear seasonal fluctuations alongside a sharp structural break during the COVID-19 period in 2020. Energy losses fell sharply due to lockdowns, reduced commuting and lower transportation activity, consistent with global reductions in transportation energy demand during the pandemic (Patino-Artaza et al., 2024). Although the series begins to recover after 2021, it does not fully return to pre-pandemic levels, suggesting a lasting reduction in baseline demand and a potential structural shift in transportation energy usage.

The STL decomposition supports these findings by showing a stable and recurring yearly seasonal pattern, indicating that while the overall level of the series changed significantly during COVID-19, the underlying seasonal structure remained largely unaffected.

2.2 Seasonal Plot

```
ggseasonplot(ts_ElecSystemLoss) +  
  ggtitle("Dataset 4: Seasonal Pattern (2010–2025)") +  
  xlab("Month") +  
  ylab("Energy Losses (Trillion Btu)")
```

→ **ggseasonplot()**: visualises recurring seasonal patterns across months and



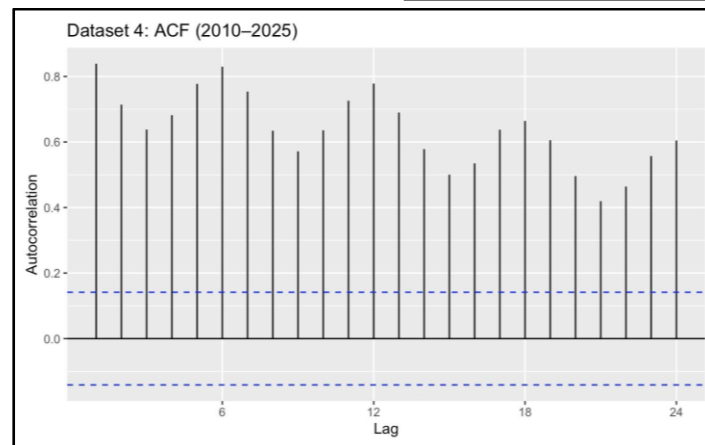
The seasonal plot confirms a strong and stable yearly seasonal pattern in transportation-related electrical energy losses, with recurring peaks typically occurring around July and December and lower values observed in April. Earlier years, particularly 2010-2013, exhibit higher seasonal peaks compared to the post-2020 period, indicating a reduction in overall demand levels over time.

Next, the 2020 observations stand out as a clear anomaly, with a sharp decline in energy losses due to COVID-19 lockdowns and mobility restrictions Kaneko et al. (2024). Although the series gradually recovered after 2021, seasonal peaks remain below pre-pandemic levels, suggesting a lasting structural shift in transportation energy consumption behaviour.

2.3 ACF Plot

```
ggAcf(ts_ElecSystemLoss) +
  ggtitle("Dataset 4: ACF (2010–2025)") +
  xlab("Lag") +
  ylab("Autocorrelation")
```

→ **ggAcf()**: used to examine autocorrelation and seasonal



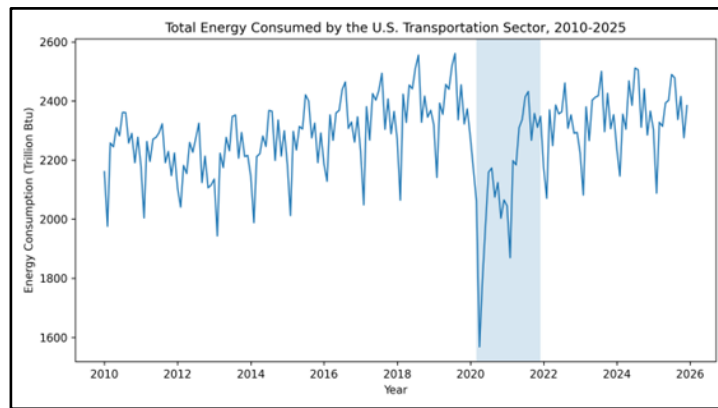
The ACF plot shows strong positive autocorrelation with a slow decay pattern, indicating that the series is non-stationary and influenced by underlying trend behaviour. Significant spikes at lag 12 and lag 24 further confirm strong annual seasonal dependence within the series. Although the COVID-19 disruption in 2020 introduced a noticeable downward level shift, the overall autocorrelation structure remained largely consistent. Therefore, both first and seasonal differencing were required to achieve stationarity before ARIMA modelling and forecasting.

DATASET 5: Total Energy Consumed by The Transportation Sector (TEABUS)

2.1 Time Series Plot

```
transport_ts <- ts_TotalEnergy
p1 <- ggplot(transport_data, aes(x = Date, y = Value)) +
  geom_rect(aes(xmin = as.Date("2020-03-01"), xmax = as.Date("2021-12-01"),
    ymin = -Inf, ymax = Inf), inherit.aes = FALSE, alpha = 0.18) +
  geom_line(linewidth = 0.8)
ggseasonplot(transport_ts, year.labels = TRUE, year.labels.left = TRUE)
transport_stl <- stl(transport_ts, s.window = "periodic")
autoplot(transport_stl)
```

→ **ggplot()**: visualises monthly U.S. transportation energy consumption overtime with a shaded region



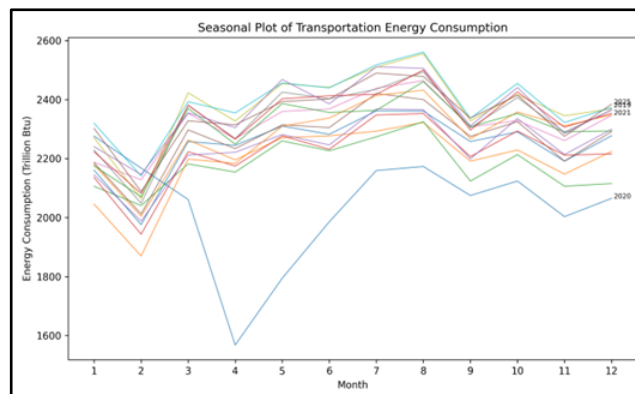
Before 2020, the series fluctuates around a relatively stable long-run level with recurring monthly peaks and troughs. The shaded COVID period begins in March 2020, and the plot shows a sudden drop immediately after that boundary. This movement is much sharper than the pre-2020 seasonal oscillations. The visual evidence therefore supports treating the early pandemic period as a structural break: the decline is not simply a normal low month, but a large negative shock connected to reduced travel, commuting, and transportation activity.

2.2 Seasonal Plot

```
# Stage 2.2 Seasonal plot
p_seasonal <- ggseasonplot(transport_ts,
  year.labels = TRUE,
  year.labels.left = TRUE) +
  labs(title = "Seasonal Plot of Transportation Energy Consumption",
    x = "Month", y = paste("Energy Consumption (", unit_label, ")"), sep = "") +
  theme_minimal()

ggsave("figures/stage2_fig2_seasonal_TotalEnergy.png", p_seasonal, width = 8, height = 5.2, dpi = 300)
```

→ **ggseasonplot()**: visualises recurring seasonal patterns across months and

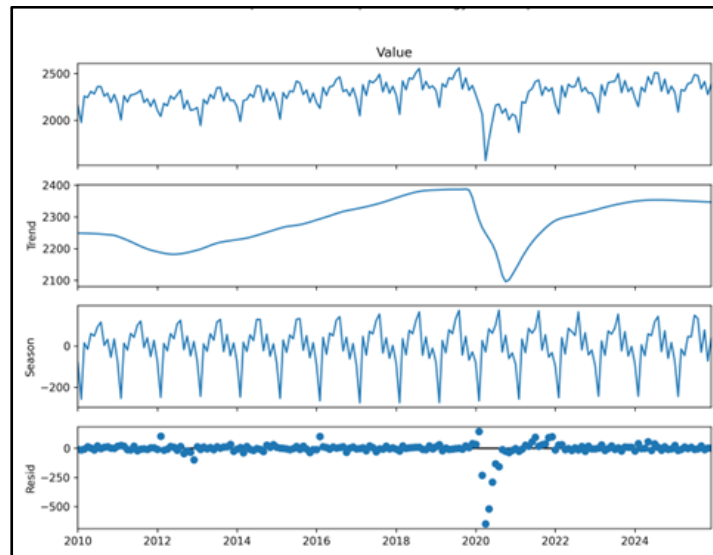


The seasonal plot compares the same calendar months across different years. Most years follow a similar within-year pattern: consumption tends to be lower in the early part of the year and higher around late spring and summer. The 2020 line is an exception because it drops far below the usual range around the early pandemic months, especially near April. This tells us two things. First, the series has annual seasonality, so the later ARIMA models need a seasonal period of 12. Second, the COVID decline is not explained by regular seasonality alone, because the 2020 path falls outside the normal seasonal envelope.

2.3 STL Decomposition

```
# Stage 2.3 STL decomposition
transport_stl <- stl(transport_ts, s.window = "periodic")
p_stl <- autoplot(transport_stl) +
  labs(title = "STL Decomposition of Transportation Energy Consumption") +
  theme_minimal()
ggsave("figures/stage2_fig3_stl_TotalEnergy.png", p_stl, width = 8, height = 6, dpi = 300)
```

→ **stl()**: applied to decompose the data into trend, seasonal and irregular



The STL decomposition separates the observed series into data, trend, seasonal, and remainder components. The trend component rises gradually before 2020, then falls sharply during the pandemic period before recovering. The seasonal component remains stable and repetitive, which confirms that monthly seasonality is a persistent feature of the series. The remainder component contains unusually large negative observations around 2020. This is the key diagnostic: if the COVID movement were ordinary seasonality, it would mainly appear in the seasonal component; instead, it appears as an abnormal remainder shock. Numerically, the 2020 monthly mean is 14.71% lower than the 2019 monthly mean, which gives a simple magnitude check for the disruption.

Measure	Value
2019 monthly mean	2389.130
2020 monthly mean	2037.724
Percentage change: 2020 vs 2019	-14.709

Stage 3: Partition Series

```
split_list <- lapply(dataset_list, function(ts_data) {
  train <- window(ts_data, end = c(2020, 2))
  test <- window(ts_data, start = c(2020, 3))

  list(
    train = train,
    test = test
  )
})

## TABLE OUTPUT ####
split_table <- do.call(rbind, lapply(names(dataset_list), function(name) {
  train_ts <- get(paste0("train_", name))
  test_ts <- get(paste0("test_", name))

  train_len <- length(train_ts)
  test_len <- length(test_ts)
  total_len <- train_len + test_len

  data.frame(
    Dataset = name,
    Period = paste(
      paste(start(train_ts), collapse = "-"),
      "to",
      paste(end(test_ts), collapse = "-")
    ),
    Split_Portion = paste0(
      round(train_len/total_len*100), "% / ",
      round(test_len/total_len*100), "%"
    ),
    Training_set_length = train_len,
    Test_set_length = test_len
  )
}))

knitr::kable(split_table,
  caption = "Train-Test Split Summary")
```

→ **lapply()**: applies the same train-test split procedure across all time series datasets automatically

→ **window()**: partitions each time series chronologically into a training set (Jan 2010 - Feb 2020) and a testing set (Mar 2020 onward)

Dataset	Period	Split_Portion	Training_set_length	Test_set_length
TotalPrimaryEnergy_Trans	2010-1 to 2025-12	64% / 36%	122	70
ElecSales	2010-1 to 2025-12	64% / 36%	122	70
EndUseEnergy	2010-1 to 2025-12	64% / 36%	122	70
ElecSystemLoss	2010-1 to 2025-12	64% / 36%	122	70
TotalEnergy	2010-1 to 2025-12	64% / 36%	122	70
TotalPrimaryEnergy_End	2010-1 to 2025-12	64% / 36%	122	70

Instead of an 80/20 split, the data is divided chronologically, with January 2010-February 2020 used for training and March 2020 onward reserved for testing. This ensures that the model is trained entirely on pre-COVID data while allowing the testing period to evaluate model performance under the structural disruptions introduced by the pandemic, even though the split ratio exceeds the typical split ratio.

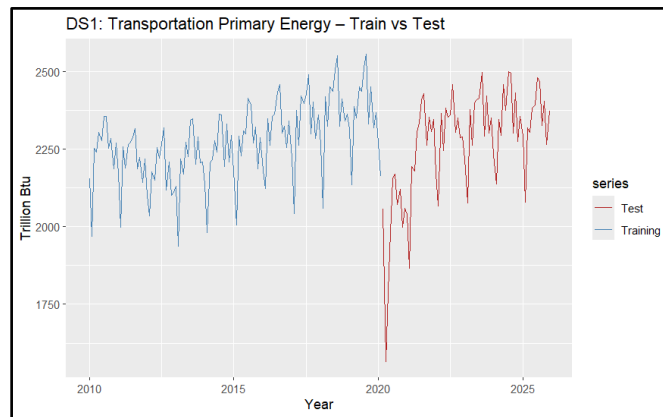
Stage 4: Prepare Data for ARIMA analysis

DATASET 1: Total Primary Energy Consumed by the Transportation Sector (TXACBUS)

4.1 Visualise Train vs Test Plot

```
autoplot(train_TotalPrimaryEnergy_Trans, series = "Training") +
  autolayer(test_TotalPrimaryEnergy_Trans, series = "Test") +
  ggtitle("DS1: Transportation Primary Energy - Train vs Test") +
  xlab("Year") + ylab("Trillion Btu") +
  scale_colour_manual(values = c("Training" = "steelblue", "Test" = "firebrick"))
```

→ **autoplot()** & **autolayer()**: Plot the training and test sets in



The training set from 2010 to 2020 shows a steady upward trend and regular seasonal patterns. The test set, covering 2020 to 2025, includes the COVID-19 decline and the recovery that followed.

4.2 BoxCox lambda transformation

```
# Check BoxCox lambda to justify transformation
lambda_Trans <- BoxCox.lambda(train_TotalPrimaryEnergy_Trans)
cat("BoxCox Lambda:", lambda_Trans, "\n")
```

The Box-Cox lambda was estimated at -0.8342, which is close to -1, showing highly unstable variance that grows steeply with the level of the series.

4.3 Differencing Requirements

```
cat("Suggested regular differences :", ndiffs(train_TotalPrimaryEnergy_Trans), "\n")
cat("Suggested seasonal differences :", nsdiffs(train_TotalPrimaryEnergy_Trans), "\n")
```

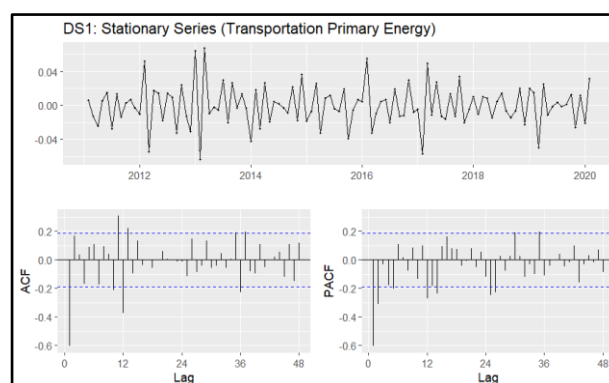
→ **ndiffs()**: Determines the number of regular difference (d) required for

4.4 Transformations & Stationary Series Plot

```
# Apply transformations to achieve stationarity
stationary_Trans <- train_TotalPrimaryEnergy_Trans %>%
  log() %>%
  diff(lag = 12) %>% # seasonal difference (D=1 at lag 12)
  diff(lag = 1) %>% # regular first difference (d=1)

# Plot stationary series with ACF and PACF
ggtsdisplay(stationary_Trans, lag.max = 48,
  main = "DS1: Stationary Series (Transportation Primary Energy)")
```

→ **log()**: Stabilises the variance of the training series



The transformed series fluctuates randomly around zero, confirming that stationarity has been successfully achieved after applying logarithmic transformation together with first and

seasonal differencing. This indicates that both the trend and seasonal effects have been adequately removed, making the series suitable for ARIMA modelling.

Next, the ACF plot exhibits a significant spike at lag 12, indicating the presence of a seasonal MA with $Q = 1$. The remaining lags are largely insignificant, suggesting limited evidence of additional non-seasonal MA dependence within the series. In contrast, the PACF plot shows a strong negative spike at lag 1 followed by smaller spikes at subsequent lags and around lag 12, providing evidence of both non-seasonal and seasonal autoregressive behaviour. This suggests plausible autoregressive orders of $p = 1-2$ and a seasonal AR component with $p = 1$.

Thus, the ACF and PACF diagnostics suggest that seasonal ARIMA specifications such as $ARIMA(1-2,1,0)(1,1,1)[12]$ and $ARIMA(0,1,1)(0,1,1)[12]$ are appropriate initial candidate models for further estimation and model evaluation.

4.5 KPSS & ADF Test

```
# KPSS Test (H0: stationary) - want p > 0.05
cat("\n--- KPSS Test: DSI Transportation Primary Energy ---\n")
print(summary(ur.kpss(stationary_Trans)))

# ADF Test (H0: unit root) - want p < 0.05
cat("\n--- ADF Test: DSI Transportation Primary Energy ---\n")
print(summary(ur.df(stationary_Trans, type = "drift", selectlags = "AIC")))
```

→ **ur.kpss()**: Test the null hypothesis that the series is stationary

```
#####
# Augmented Dickey-Fuller Test Unit Root Test #
#####

Test regression drift

Call:
lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)

Residuals:
    Min       1Q   Median       3Q      Max
-0.06304 -0.01052 -0.00050  0.01104  0.06119

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0001756   0.0017549  -0.100   0.92049
z.lag.1      -2.1238747   0.1677493 -12.661 < 2e-16 ***
z.diff.lag    0.3172170   0.0937237   3.385   0.00101 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01815 on 104 degrees of freedom
Multiple R-squared:  0.8233, Adjusted R-squared:  0.8199
F-statistic: 242.3 on 2 and 104 DF, p-value: < 2.2e-16
```

Value of test-statistic is: -12.661 80.1586

Critical values for test statistics:

	1pct	5pct	10pct
tau2	-3.46	-2.88	-2.57
phi1	6.52	4.63	3.81

```
#####
# KPSS Unit Root Test #
#####

Test is of type: mu with 4 lags.

Value of test-statistic is: 0.0356

Critical value for a significance level of:
      10pct   5pct  2.5pct  1pct
critical values 0.347 0.463 0.574 0.739
```

Test	H_0	Test Statistics	Critical Values (5%)	Decision
KPSS	Stationary	0.0356	0.463	Fail to reject H_0
ADF	Unit root exists	-12.661	-2.88	Reject H_0

Conclusion: Both tests agree that the transformed series is stationary and ready for ARIMA model fitting.

DATASET 2: Electricity Sales to Ultimate Customers in the Transportation Sector (ESACBUS)

```
{r}
# plotting training and testing dataset on a single plot
autoplot(train_ElecSales, series = "Training Data") +
  autoplot(test_ElecSales, series = "Test Data") +
  labs(title = "Training and Test Dataset Split", x = "Year", y = "Value") +
  theme_minimal()

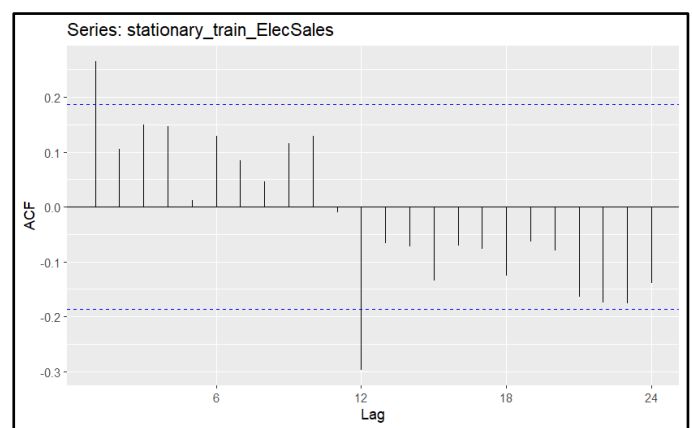
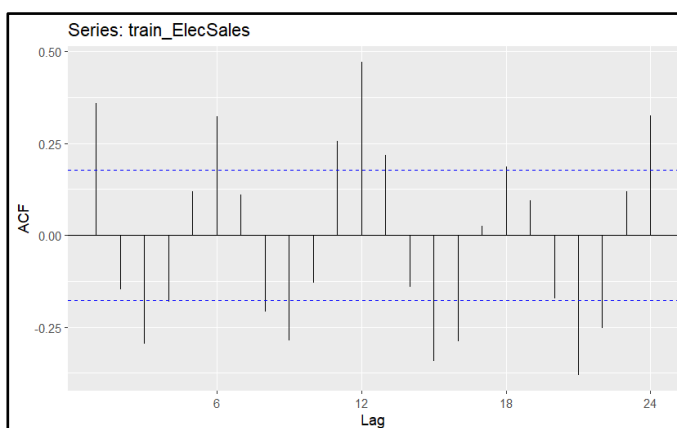
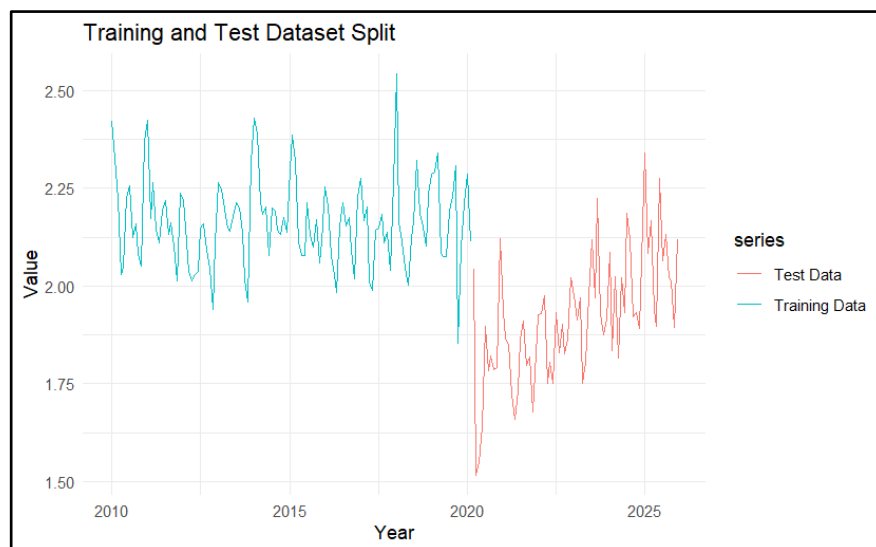
#these are the tests and acf plot to prove that the data is stationary
adf.test(train_ElecSales)
kpss.test(train_ElecSales)
ggAcf(train_ElecSales)

#transforming data using differencing
stationary_train_ElecSales <- diff(train_ElecSales, lag = 12)
adf.test(stationary_train_ElecSales)
kpss.test(stationary_train_ElecSales)
ggAcf(stationary_train_ElecSales)
```

→ **ADF test:** Tests for unit root (non-stationarity) in the series

→ **KPSS test:** Tests for stationarity as the null hypothesis

This code is used to prepare the data for modelling. It starts by visualizing the data split to ensure the patterns are consistent. Then, it uses ADF and KPSS tests along with an ACF plot to check for stationarity and seasonal patterns. Finally, it applies a seasonal difference (lag = 12) to see if removing the yearly dependency makes the data better for the AR and MA models. The training series exhibited approximately constant variance across time; therefore, a Box-Cox transformation was deemed unnecessary.



The initial analysis of the training data (train_ElecSales) provided strong statistical evidence of stationarity, with the ADF test yielding a p-value of 0.01 and the KPSS test a p-value of 0.1. Despite these results, the ACF plot—which is built to determine the extent to which current data is dependent on values from previous years and months—revealed significant spikes at 12-month intervals. This suggests the presence of seasonal non-stationarity that the standard tests did not fully capture.

To address this seasonal memory, an experimental seasonal difference (lag = 12) was applied. However, the results for the transformed dataset (stationary_train_ElecSales) were less favorable. The ADF p-value increased to 0.1317, failing the requirement for stationarity at the 5% significance level. Furthermore, the ACF plot for the differenced data continued to show a dominant spike at Lag 12, indicating that the series remained heavily dependent on specific yearly patterns.

The deterioration of the test statistics suggests that the transformation led to over-differencing, adding unnecessary complexity without resolving the seasonal dependency. Consequently, the original non-differenced data was retained for the construction of the AR and MA models. This approach ensures the models utilize a dataset that is statistically stationary while preserving the underlying structure necessary for accurate forecasting.

DATASET 3 : End-use energy consumption in the transportation sector (TNACBUS)

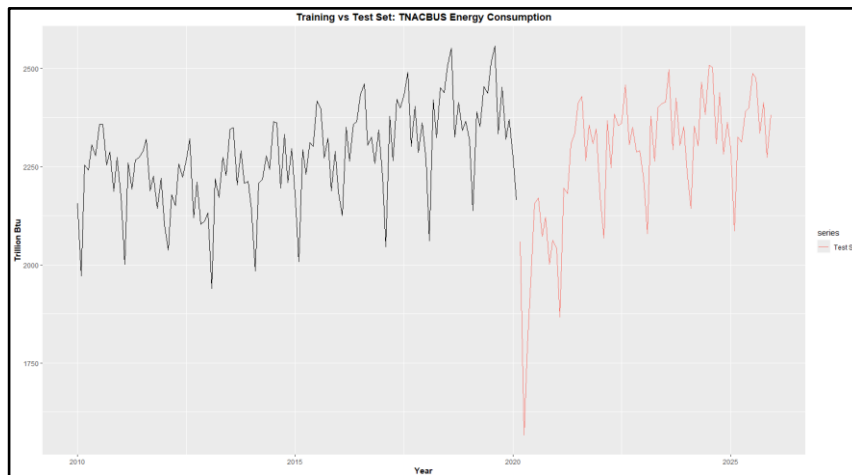
4.1 Visualisation of Training VS Test Set

```
# STAGE 4: Prepare Data for ARIMA analysis ####
TNACBUS_train <- split_list$EnduseEnergy$train
TNACBUS_test <- split_list$EnduseEnergy$test

## 4.1 Visualise Training VS Test Set ####
library(tseries)

# Plot Training vs Test Set
autoplot(TNACBUS_train) +
  autolayer(TNACBUS_test, series = "Test Set") +
  ggtitle("Training vs Test Set: TNACBUS Energy Consumption") +
  xlab("Year") +
  ylab("Trillion Btu") +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.title = element_text(face = "bold")
  )
```

→ **autolayer()**: Overlays the test set on the same graph to enable direct visual comparison



From the plot, the training set shows relatively stable transportation energy demand before the pandemic, while the test set contains a sharp decline during COVID-19 followed by gradual recovery. This indicates that the post-pandemic period behaves differently from the historical period, making forecasting more challenging due to sudden structural changes in demand.

4.2 Stationarity Assessment of Original Training Data (Before Transformation)

```
> ### 4.2.1 ACF of original training data ###
> ggAcf(TNACBUS_train) +
+   theme(
+     plot.title = element_text(face = "bold", hjust = 0.5),
+     axis.title = element_text(face = "bold")
+   )
> ### 4.2.2 KPSS test (stationarity check) ###
> kpss.test(TNACBUS_train)

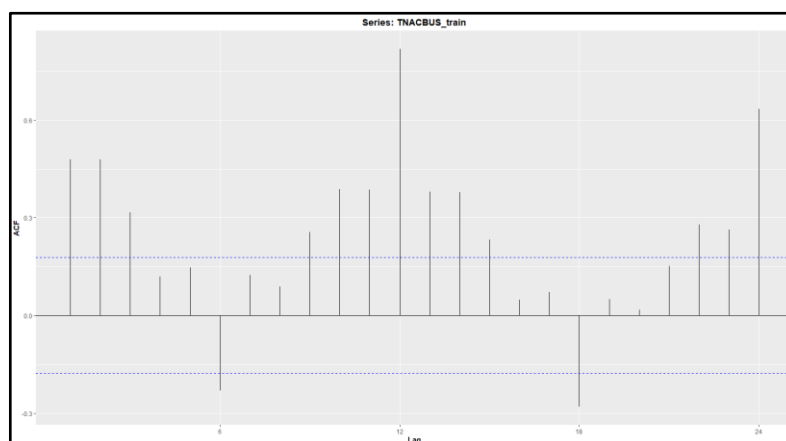
    KPSS Test for Level Stationarity

data:  TNACBUS_train
KPSS Level = 1.3197, Truncation lag parameter = 4, p-value = 0.01

warning message:
In kpss.test(TNACBUS_train) : p-value smaller than printed p-value
```

→ **ggAcf()**: Examines autocorrelation between current observations and lagged values

• **kpss.test()**: Test whether the time series is



The ACF plot shows strong autocorrelation and seasonal spikes at lag 12, indicating the presence of trend and seasonality in the series. Next, the KPSS test further confirms this result, with a test statistic of 1.1923 and a p-value below 0.05. This shows that the KPSS test rejects

the null hypothesis of stationarity, confirming that the original training dataset for TNACBUS is non-stationary.

4.3 Data Transformation (Make Stationary)

```
### 4.3.1 Log transformation (variance stabilisation) ###
train_log <- log(TNACBUS_train)

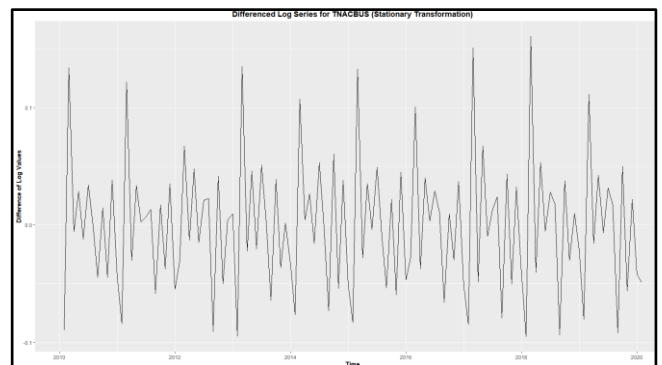
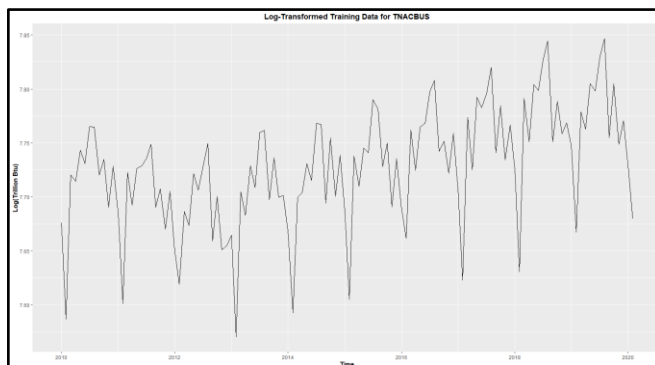
autoplot(train_log) +
  ggtitle("Log-Transformed Training Data") +
  ylab("Log(Trillion $tu)") +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.title = element_text(face = "bold")
  )

### 4.3.2 First differencing (trend removal) ###
train_diff <- diff(train_log)

autoplot(train_diff) +
  ggtitle("Differenced Log Series (Stationary Transformation)") +
  ylab("Difference of Log Values") +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.title = element_text(face = "bold")
  )
```

→ **log()**: Applies logarithmic transformation to stabilise the variance of the series

→ **autoplot()**: Visualises the transformed time series data



Based on the above plot, the log-transformed series displays a more stable variance compared to the original TNACBUS dataset. The log transformation compresses large values and reduces heteroscedasticity, making the variance more stable across time. As a result, extreme fluctuations, especially during the COVID-19 period, are reduced and become smoother. However, trend movements are still present, indicating that differencing is required to achieve stationarity.

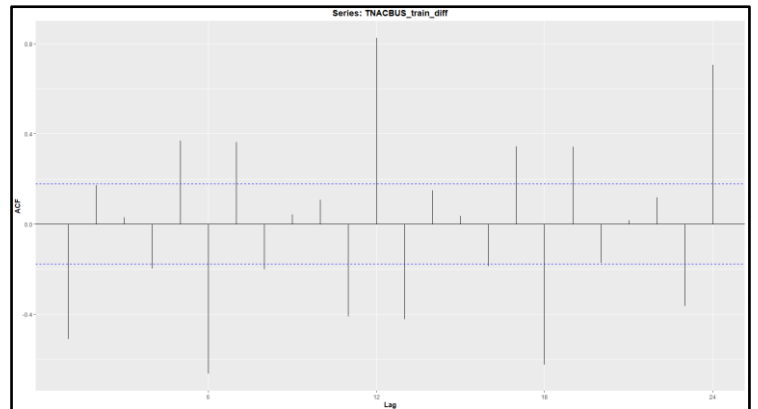
First differencing is applied to the log-transformed series to remove the trend component. After differencing, the TNACBUS transformed series fluctuates around a constant mean with no obvious long-term upward or downward movement. This indicates that the differencing process has successfully removed the trend component that was present in the original data and successfully improved the stationarity of the series.

4.4 Post-Transformation Diagnostics (After Transformation)

```
> ## 4.4 Check Transformed Data (After Transformation) ####
> ### 4.4.1 ACF after transformation ###
> ggacf(TNACBUS_train_diff) +
+ theme(
+   plot.title = element_text(face = "bold", hjust = 0.5),
+   axis.title = element_text(face = "bold")
+ )
> ### 4.4.2 KPSS test after transformation ###
> kpss.test(TNACBUS_train_diff)

KPSS Test for Level Stationarity
data: TNACBUS_train_diff
KPSS Level = 0.042991, Truncation lag parameter = 4, p-value = 0.1

warning message:
In kpss.test(TNACBUS_train_diff) : p-value greater than printed p-value
```



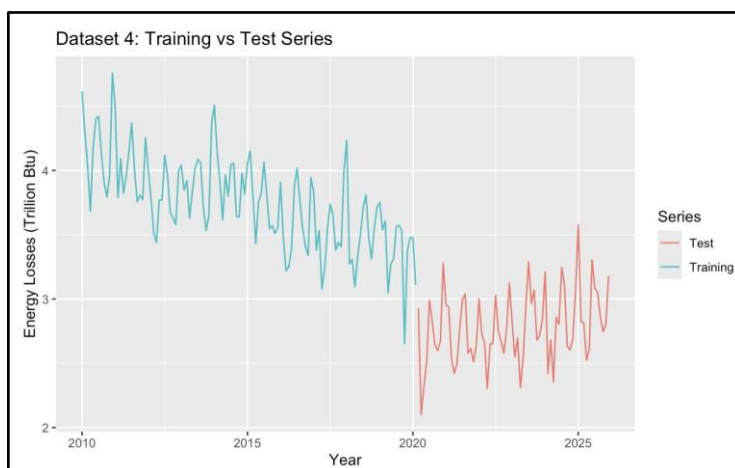
After transformation, most autocorrelation values fall within the confidence bounds, indicating that long-term dependence has been substantially reduced. In addition, the KPSS test gives a p-value greater than 0.05, demonstrating that we fail to reject the null hypothesis of stationarity. This confirms that the combination of log transformation and first differencing successfully stabilised the series and prepared the data for ARIMA modelling.

DATASET 4: Electrical System Energy Losses Proportioned to the Transportation Sector (LOACBUS)

4.1 Train vs Test Plot

```
# 4.1 Training vs Test Plot
autoplot(train_ElecSystemLoss, series = "Training") +
  autolayer(test_ElecSystemLoss, series = "Test") +
  ggtitle("Dataset 4: Training vs Test Series") +
  xlab("Year") +
  ylab("Energy Losses (Trillion Btu)") +
  guides(colour = guide_legend(title = "Series"))
```

→ **autoplot()** & **autolayer()**: Plot the training and test sets in



The training series shows clear seasonal fluctuations before 2020, while the test series captures the sharp COVID-19 decline and partial recovery in transportation-related electricity demand.

4.2 Transformation & Differencing

```
# 4.3 Differencing Requirement
ndiffs(train_ElecSystemLoss)
nsdiffs(train_ElecSystemLoss)
```

→ **ndiffs()**: Determines the number of regular difference (d) required for stationarity

The estimated Box-Cox lambda value of 0.919 indicates that no transformation was required.

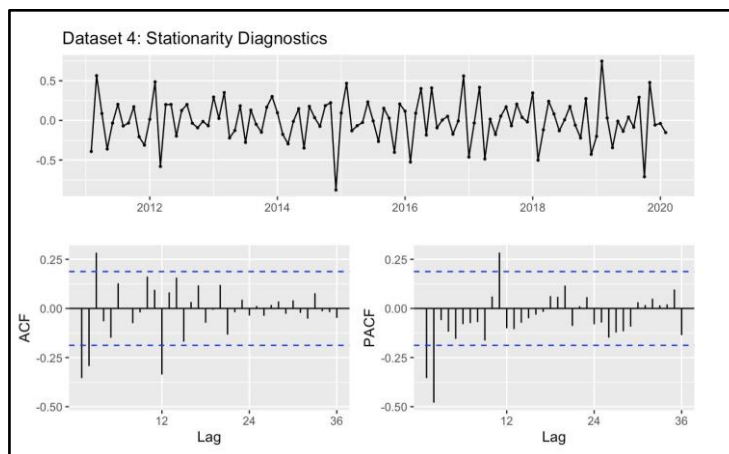
4.3 Stationary Series, ACF & PACF

```
# 4.4 Seasonal + First Differencing
ElecSystemLoss_stationary <- train_ElecSystemLoss %>%
  diff(lag = 12) %>%
  diff()

# 4.5 Stationarity Diagnostics
ggtsdisplay(ElecSystemLoss_stationary,
  main = "Dataset 4: Stationarity Diagnostics")
```

→ **diff(lag = 1)**: Applies first differencing to remove trend and achieve stationarity

→ **diff(lag = 12)**: Applies seasonal



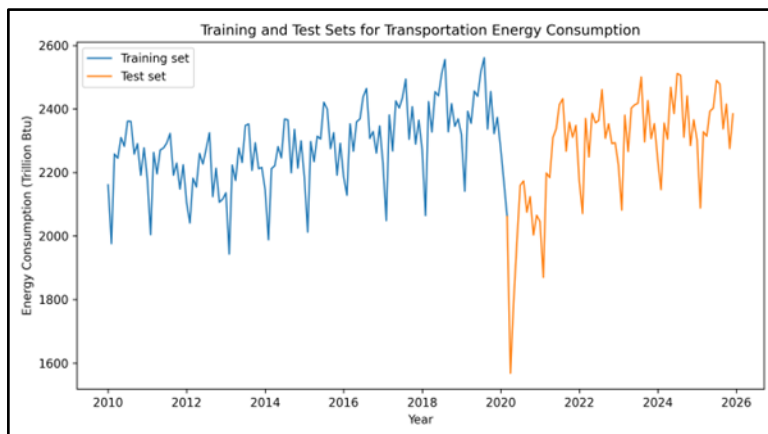
After seasonal differencing at lag 12 and first differencing, the transformed series fluctuated around a constant mean with no obvious trend, indicating that stationarity had been successfully achieved. The ACF plot showed several significant negative spikes at early lags together with a remaining seasonal spike near lag 12, suggesting the presence of short-term and seasonal moving average behaviour. Similarly, the PACF plot displayed strong spikes at the first few lags followed by a rapid decline, indicating possible autoregressive effects and a seasonal AR component. Overall, the combined ACF and PACF diagnostics suggested that seasonal ARIMA models such as $\text{ARIMA}(2,1,0)(1,1,0)[12]$ and $\text{ARIMA}(0,1,1)(0,1,1)[12]$ were reasonable candidate models for forecasting electricity system losses. In addition, the ADF test rejected the presence of a unit root while the KPSS test failed to reject stationarity, further confirming that the differenced series was suitable for ARIMA modelling.

DATASET 5: Total Energy Consumed by The Transportation Sector (TEACBUS)

The following steps are to visualize training and test sets together, apply suitable transformations, and justify stationarity. The transformation, differencing, ACF/PACF inspection, and forecast-evaluation workflow follow standard time-series practice (Hyndman & Athanasopoulos, 2021; Box et al., 2015).

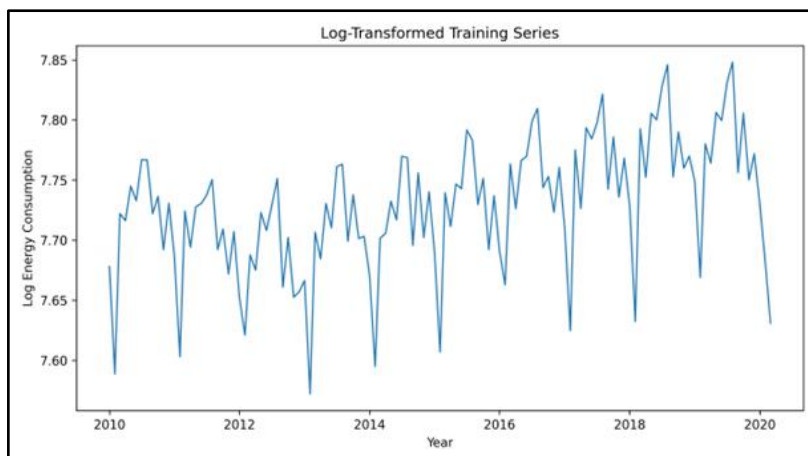
```
p4 <- autoplot(train_ts, series = "Training set") +  
  autolayer(test_ts, series = "Test set")  
  
train_log <- log(train_ts)  
train_log_sdiff <- diff(train_log, lag = 12)  
train_log_sdiff_diff <- diff(train_log_sdiff, differences = 1)  
final_train <- train_log_sdiff_diff
```

4.1 Comparison of Training and Test Sets



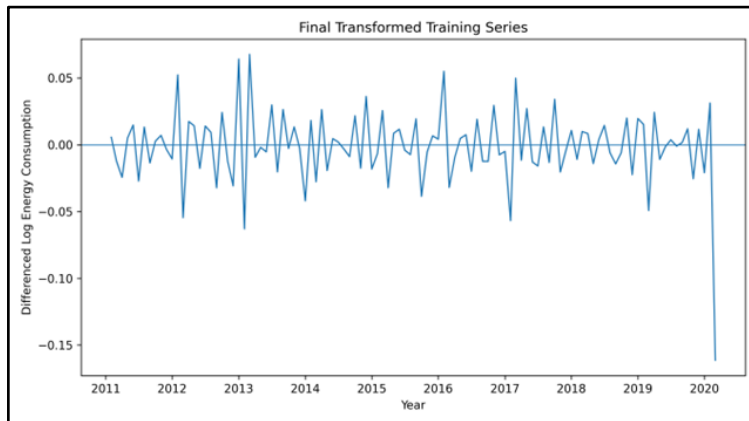
The plot separates the pre-COVID training pattern from the pandemic and post-pandemic test period. Since March 2020 is the boundary month, the test period contains the sudden COVID drop and the subsequent recovery. This setup directly matches the project aim of examining COVID effects.

4.2 Log-transformed Training Series



The log transformation is appropriate because energy consumption is strictly positive. It helps stabilize variance and makes proportional changes easier to interpret. However, the log-transformed series still shows seasonal and non-stationary behaviour, so differencing is still required.

4.3 Final Transformed Training Series

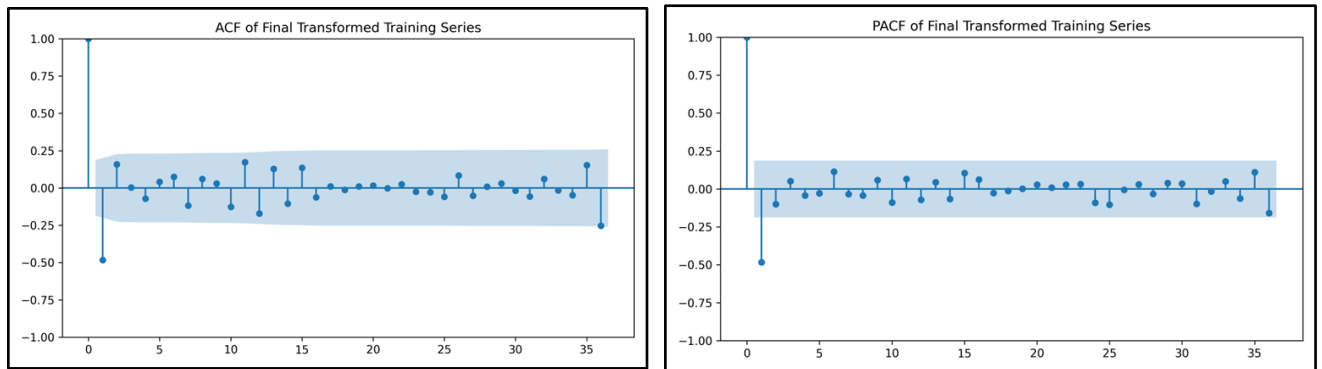


The transformed series fluctuates around a more stable mean and no longer displays the strong annual pattern visible in the raw data. This supports the use of the transformed series for ARIMA identification.

Series	ADF statistic	5% critical value	Decision
Original training series	-0.312	-2.888	Not stationary at 5%
Log training series	-0.322	-2.888	Not stationary at 5%
Log + seasonal difference	-1.798	-2.889	Not stationary at 5%
Log + seasonal difference + first difference	-10.451	-2.889	Stationary at 5%

The original and log-transformed training series are not stationary at the 5% level. Seasonal differencing alone is not sufficient. After adding one first difference, the ADF statistic becomes much smaller than the 5% critical value, so the final transformed series is treated as stationary. The stationarity decision is based on the augmented Dickey-Fuller unit-root framework (Dickey & Fuller, 1979).

4.4 ACF & PACF of Final Transformed Training Series.



ADF Stationarity Test:

Series	ADF statistic	5% critical value	Decision
Original training series	-0.312	-2.888	Not stationary
Log training series	-0.322	-2.888	Not stationary
Log + seasonal difference	-1.798	-2.889	Not stationary
Log + seasonal difference + first difference	-10.451	-2.889	Stationary

The ADF results show why both differences are needed. Logging alone does not remove non-stationarity, and one seasonal difference is still insufficient. After one additional first difference, the ADF statistic is far below the 5% critical value, so the transformed training series is treated as stationary. The Stage 5 pure AR and pure MA candidates therefore use $d = 1$, $D = 1$, and monthly seasonal period 12. The ACF and PACF then guide whether short-run dependence is better represented by AR terms or MA terms. The first-lag behaviour is still visible after transformation, but most later lags are inside the significance bands, so low-order AR and MA benchmarks are reasonable. The large COVID observation is not removed by differencing; it remains a genuine shock that later models can diagnose through residual plots and forecast errors. The modelling implication is that the ARIMA orders should be selected conservatively: the differencing removes trend and annual seasonality, while the remaining ACF/PACF pattern only justifies low-order short-memory terms. This is why Stage 5 reports deliberately restricted pure AR and pure MA benchmarks instead of overfitting many AR and MA parameters. The later comparison also separates in-sample adequacy from forecast performance, because a model can have a good AICc yet still perform poorly when the post-COVID evaluation period differs from the pre-COVID training regime.

Stage 5: Build your pure AR and MA models

DATASET 1: Total Primary Energy Consumed by the Transportation Sector (TXACBUS)

Based on the ACF and PACF analysis conducted in Stage 4, the PACF cutoff at lag 2 and seasonal spike at lag 12 justify a pure AR model of $ARIMA(2,1,0)(1,1,0)[12]$, while the ACF spike at lag 12 and cutoff at lag 1 justify a pure MA model of $ARIMA(0,1,2)(0,1,1)[12]$. These two models are fitted and evaluated below.

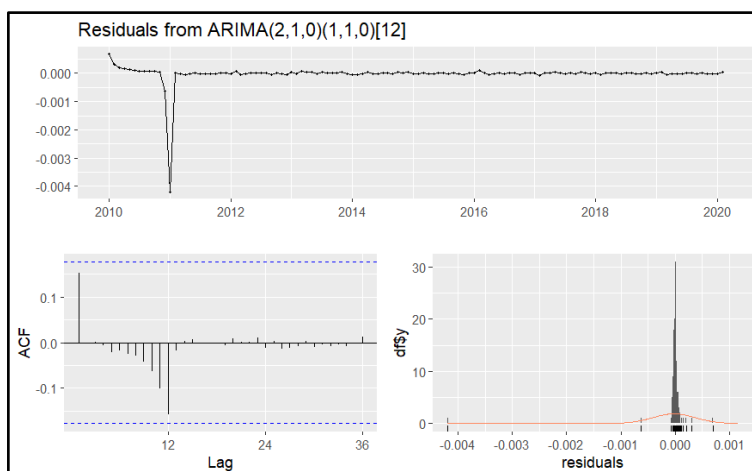
5.1 Pure AR Model

5.1.1 $ARIMA(2,1,0)(1,1,0)[12]$

```
ar_Trans <- Arima(log(train_TotalPrimaryEnergy_Trans),  
                  order = c(2, 1, 0),  
                  seasonal = list(order = c(1, 1, 0), period = 12),  
                  include.constant = TRUE)  
summary(ar_Trans)  
checkresiduals(ar_Trans)
```

→ **Arima()**: Fits an $ARIMA(2,1,0)(1,1,0)[12]$ model to the log-transformed training series

→ **log()**: Applies log transformation to



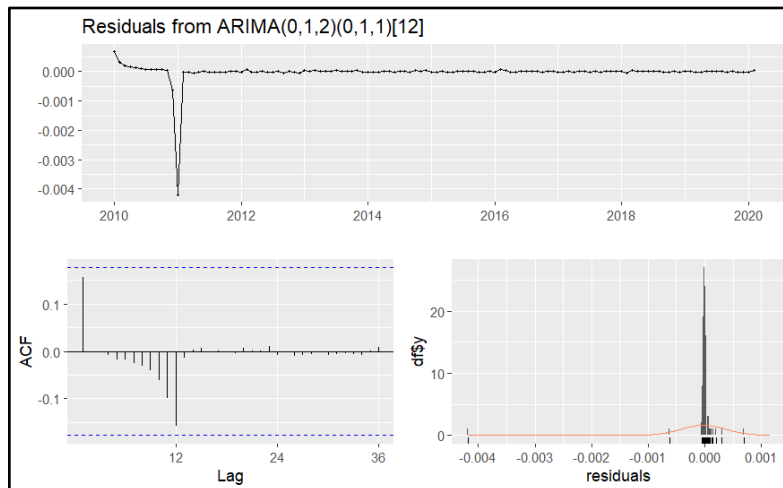
Ljung-Box test

data: Residuals from $ARIMA(2,1,0)(1,1,0)[12]$
 $Q^* = 8.9486$, $df = 21$, $p\text{-value} = 0.9896$

Model $df = 3$. Total lags used: 24

The residual plot fluctuates randomly around zero with no evident pattern, although a notable spike appears around 2011, likely reflecting an external structural shock. Next, the ACF plot shows no significant autocorrelation at any lag, as all spikes lie within the confidence bounds. The histogram confirms that the residuals are approximately normally distributed and centred around zero. In addition, the Ljung-Box test produces a high p-value of 0.9896, showing above the 0.05 threshold, confirming that the residuals are consistent with white noise and no significant autocorrelation remains.

5.1.2 ARIMA(0,1,2)(0,1,1)[12]



Ljung-Box test

data: Residuals from ARIMA(0,1,2)(0,1,1)[12]
 $Q^* = 8.9178$, $df = 21$, $p\text{-value} = 0.9898$

Model df: 3. Total lags used: 24

The residual plot fluctuates around zero with no clear trend or systematic pattern, except for an isolated spike around 2011, driven by an external structural shock. Next, the ACF plot confirms no significant autocorrelation remains, as all spikes lie within the confidence bounds. The histogram shows residuals are approximately normally distributed and centred around zero. Furthermore, the Ljung-Box test shows a p-value of 0.9898, well above 0.05, confirming that residuals behave as white noise and the model is adequately specified.

5.2 Forecasts & Back-transform

```
# Forecast (auto back-transformed since lambda used)
fc_ar_Trans <- forecast(ar_Trans,
  h = length(test_TotalPrimaryEnergy_Trans))
```

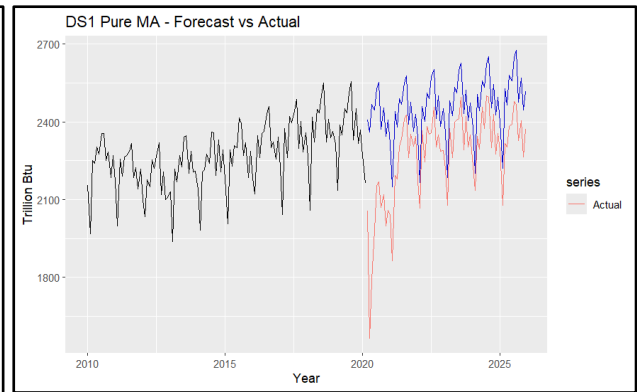
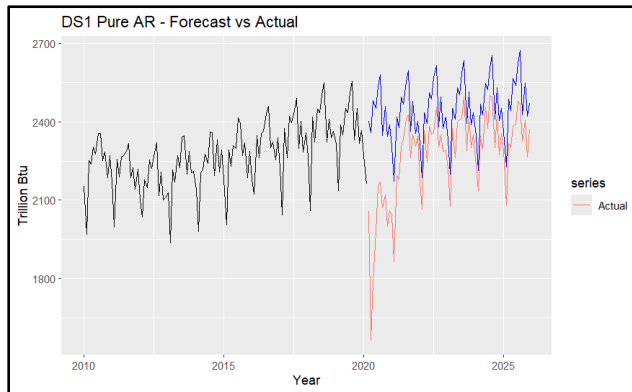
```
# Plot forecast vs actual
autoplot(fc_ar_Trans) +
  autolayer(test_TotalPrimaryEnergy_Trans, series = "Actual") +
  ggtitle("DS1 Pure AR - Forecast vs Actual") +
  xlab("Year") + ylab("Trillion Btu")
```

```
# Forecast
fc_ma_Trans <- forecast(ma_Trans,
  h = length(test_TotalPrimaryEnergy_Trans))
```

```
# Plot forecast vs actual
autoplot(fc_ma_Trans) +
  autolayer(test_TotalPrimaryEnergy_Trans, series = "Actual") +
  ggtitle("DS1 Pure MA - Forecast vs Actual") +
  xlab("Year") + ylab("Trillion Btu")
```

→ **forecast()**: Produces predictions and back-transforms them to the original scale using **exp()**

→ **autoplot()**: Plots the forecast results with historical data



Based on the above, both models fail to capture the sharp COVID-19 decline, with forecasts remaining significantly above the actual values throughout the test period. The pure AR model produces a higher and more prolonged forecast trajectory, with values remaining elevated for a longer period compared to the pure MA model. This behaviour is expected, as AR models tend to propagate past trends more strongly, sustaining momentum over time. In contrast, MA models respond more quickly to shocks and revert toward the mean at a faster rate.

5.4 Both model's AR Accuracy & Rolling Forecast Validation (tsCV)

```
# Accuracy on original scale
acc_ar_Trans <- accuracy(fc_ar_Trans$mean, test_TotalPrimaryEnergy_Trans)
print(acc_ar_Trans)

# tsCV on original scale
farima_ar <- function(x, h) {
  forecast(Arima(x,
    order = c(2, 1, 0),
    seasonal = list(order = c(1, 1, 0), period = 12),
    lambda = lambda_Trans,
    include.constant = TRUE), h = h)
}
cv_ar <- tsCV(train_TotalPrimaryEnergy_Trans, farima_ar, h = 12)
cv_ar_rmse <- sqrt(mean(cv_ar^2, na.rm = TRUE))
cv_ar_mae <- mean(abs(cv_ar), na.rm = TRUE)
cat(sprintf("AR(3)(2,1,0)[12] | tsCV RMSE: %.4f | tsCV MAE: %.4f\n",
  cv_ar_rmse, cv_ar_mae))
```

```
# Accuracy
acc_ma_Trans <- accuracy(fc_ma_Trans$mean, test_TotalPrimaryEnergy_Trans)
print(acc_ma_Trans)

# tsCV
farima_ma <- function(x, h) {
  forecast(Arima(x,
    order = c(0, 1, 2),
    seasonal = list(order = c(0, 1, 1), period = 12),
    lambda = lambda_Trans,
    include.constant = TRUE), h = h)
}
cv_ma <- tsCV(train_TotalPrimaryEnergy_Trans, farima_ma, h = 12)
cv_ma_rmse <- sqrt(mean(cv_ma^2, na.rm = TRUE))
cv_ma_mae <- mean(abs(cv_ma), na.rm = TRUE)
cat(sprintf("MA(1)(0,1,1)[12] | tsCV RMSE: %.4f | tsCV MAE: %.4f\n",
  cv_ma_rmse, cv_ma_mae))
```

→ **accuracy()**: Computes forecast accuracy measures on the test set, including RMSE and MAE

→ **tsCV()**: Performs time series cross-validation to evaluate out-of-sample performance, calculating errors such as

Model	AIC	Ljung-Box p-value	Test RMSE	Test MAE	Test MAPE (%)	tsCV RMSE	tsCV MAE
AR(2)(1,1,0)[12]	-1970.32	0.9896	219.59	176.52	8.24	47.57	37.16
MA(2)(0,1,1)[12]	-2004.65	0.9898	226.03	188.80	8.77	50.95	38.53

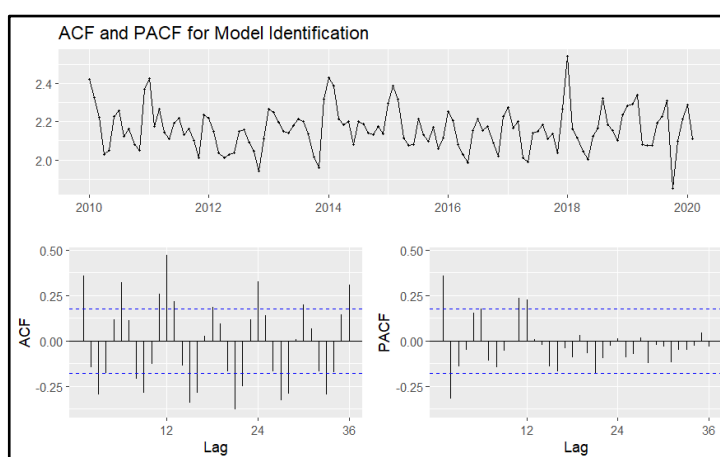
Both models pass the Ljung-Box test, confirming that the residuals behave as white noise and no significant autocorrelation remains. Although the MA(2)(0,1,1)[12] model achieved a lower AICc (-2004.65 vs -1970.32), showing a better in-sample fit, the AR(2)(1,1,0)[12] model outperformed in out-of-sample evaluation, with lower test RMSE (219.59 vs 226.03) and tsCV RMSE (47.57 vs 50.95). Based on tsCV RMSE as the primary selection criterion, the AR(2)(1,1,0)[12] model is selected as the preferred pure model. However, both models recorded Theil's U greater than 1, suggesting limited forecasting ability during the COVID-19 period.

DATASET 2: Electricity Sales to Ultimate Customers in the Transportation Sector (ESACBUS)

5.1 ARIMA Model Identification and Seasonal Structure Diagnostics

```
{r}
ggtsdisplay(train_ElecSales, main="ACF and PACF for Model Identification")
```

→ **ggtsdisplay()**: shows the



The PACF plot shows a single significant spike at lag 1 followed by a sharp cut-off, suggesting that an AR(1) term is appropriate for capturing the non-seasonal structure. Next, the ACF plot displays a significant spike at lag 1, demonstrating the presence of an MA(1) component, while strong repeating spikes at lag 12, 24 and 36 reveal pronounced seasonal dependence. This

pattern supports the inclusion of a seasonal MA(1) term to capture yearly cyclical behaviour, Although manual seasonal differencing was initially considered, it leads to signs of over-differencing and weakened stationarity based on ADF and KPSS tests. Therefore, the seasonal structure is better handled directly within the ARIMA framework, allowing the model to account for the 12-month dependency while preserving the integrity of the original series and improving overall model stability and diagnostics.

5.2 Fitting the AR and MA Models

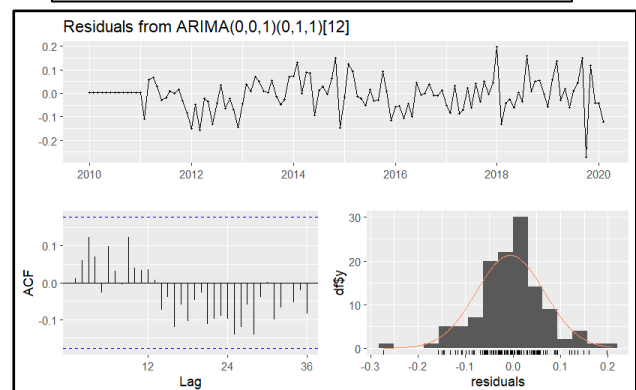
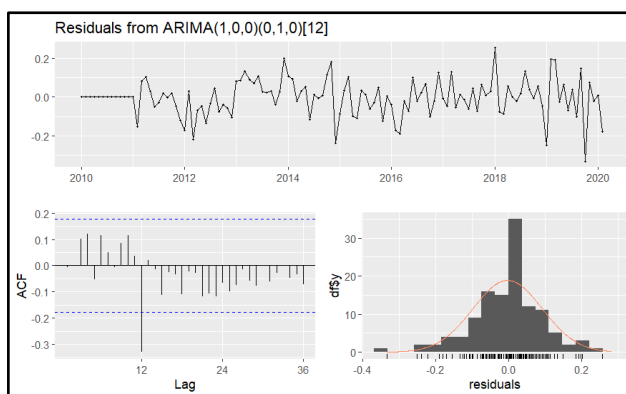
```
[r]
# This is a pure AR(1) with seasonal differencing
fit_ar <- Arima(train_ElecSales, order = c(1, 0, 0), seasonal = c(0, 1, 0))

# This is a pure MA(1) with seasonal differencing
fit_ma <- Arima(train_ElecSales, order = c(0, 0, 1), seasonal = c(0, 1, 0))

# Checking residuals
checkresiduals(fit_ar)
checkresiduals(fit_ma)
```

→ **Arima()**: Fits an ARIMA model to the log-transformed training series

→ **log()**: Applies log transformation to stabilise variance in the data



The AR model (`fit_ar`) is specified with a non-seasonal order of (1,0,0), while the MA model (`fit_ma`) is specified as (0,1,1). Both models incorporate a seasonal difference of (0,1,0) at a period of 12 to account for the annual seasonality observed in the ACF plot. For the AR model, the Ljung-Box test shows a p-value of 0.061, which is above the 0.05 significant level. This suggests that the residuals are not significantly autocorrelated and can be considered approximately white noise. However, the residual ACF still shows a noticeable negative spike at lag 12, indicating that the seasonal pattern is not fully captured by simple differencing.

In contrast, the MA model fails the diagnostics test, with a Ljung-Box p-value of 0.029, which is below 0.05. This indicates the presence of significant residual autocorrelation. The residual ACF also shows a strong spike at lag 12, confirming that the MA(1) specification combined with seasonal differencing is insufficient to fully capture the seasonal structure.

Therefore, the result suggests that while the AR model is marginally adequate, both models struggle to fully represent the seasonal dependence in the data, highlighting the need for a more flexible seasonal ARIMA specification.

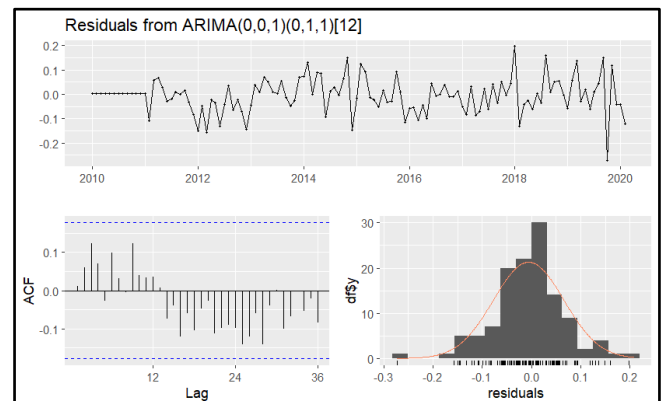
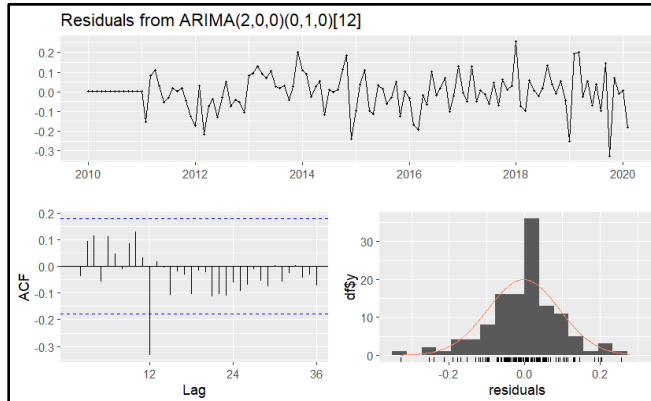
5.3 Reidentification

```
{r}
# Reidentified Pure AR model
fit_ar_reid <- Arima(train_ElecSales, order = c(2, 0, 0), seasonal = list(order = c
(0, 1, 0), period = 12))

# MA(1) with a Seasonal MA(1) term
fit_ma_reid <- Arima(train_ElecSales, order = c(0, 0, 1), seasonal = list(order = c
(0, 1, 1), period = 12))

checkresiduals(fit_ar_reid)
checkresiduals(fit_ma_reid)
```

→ **checkresiduals()**: Used to assess whether the model residuals behave like white noise and whether



The AR model was re-specified by increasing the non-seasonal order from AR(1) to AR(2), while retaining seasonal differencing at lag 12. This adjustment aims to better capture remaining short-term autocorrelation, resulting in an ARIMA(2,0,0)(0,1,0)[12] specification.

For the MA model, a seasonal MA(1) component was introduced to address persistent seasonal autocorrelation at lag 12, while maintaining the non-seasonal MA(1) structure, leading to an ARIMA(0,0,1)(0,1,1)[12] model.

The diagnostic results show clear improvement in model adequacy. The AR model produces a Ljung-Box p-value of 0.05298, while the MA model achieves 0.6578. Since both values exceed the 0.05 significance level, the residuals can be considered independent and consistent with white noise, indicating that both models adequately capture the underlying structure of the series.

5.4 Traditional Accuracy

```
{r}
# Generate forecasts
fc_ar <- forecast(fit_ar_reid, h = length(test_ElecSales))
fc_ma <- forecast(fit_ma_reid, h = length(test_ElecSales))

# Check traditional accuracy
accuracy(fc_ar, test_ElecSales)
accuracy(fc_ma, test_ElecSales)
```

→ **forecast()**: Generates predictions over the forecast horizon (h), matching the length of the test data

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	-0.002824732	0.09459145	0.06838231	-0.22757	3.157917	0.8748609	-0.002240528	NA
Test set	-0.229365655	0.28067578	0.24572273	-12.52939	13.307353	3.1436963	0.268726501	1.81054
	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	-0.005897367	0.07148976	0.05187819	-0.3620805	2.40514	0.6637126	0.01074755	NA
Test set	-0.238446448	0.27578607	0.24465219	-13.0393025	13.31381	3.1300001	0.52601090	1.791742

The accuracy results show that both models perform very similarly, with no clear dominant model across all evaluation metrics. On the test set, the MA model achieves slightly lower RMSE (0.2758) and MAE (0.2447), indicating marginally better performance in terms of absolute forecast error. However, the AR model recorded a slightly lower MAPE of 13.31%, suggesting marginally better relative percentage accuracy.

Thus, the differences in performance are minimal, indicating that both models provide comparable forecasting capability for this dataset, with no consistent superiority across all metrics.

5.5 tsCV

```
[r]
# Define forecast functions for tsCV
f_ar <- function(x, h) {forecast(Arima(x, order=c(2,0,0), seasonal=c(0,1,0)), h=h)}
f_ma <- function(x, h) {forecast(Arima(x, order=c(0,0,1), seasonal=c(0,1,1)), h=h)}

# Run tsCV (h=12)
cv_ar <- tsCV(train_ElecSales, f_ar, h=12)
cv_ma <- tsCV(train_ElecSales, f_ma, h=12)

# Calculate RMSE for the cross-validation
rmse_cv_ar <- sqrt(mean(cv_ar^2, na.rm=TRUE))
rmse_cv_ma <- sqrt(mean(cv_ma^2, na.rm=TRUE))

print(rmse_cv_ar)
print(rmse_cv_ma)
```

```
[1] 0.09556642
[1] 0.1646117
```

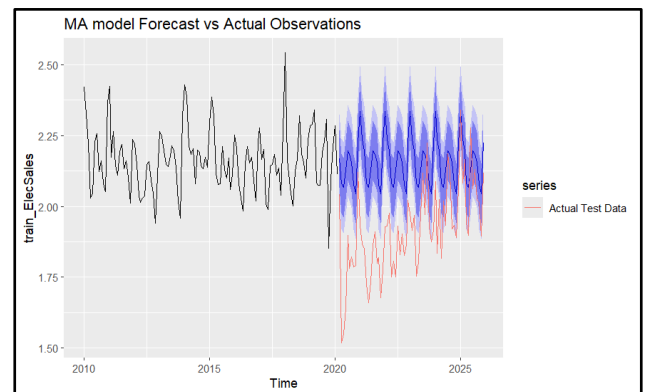
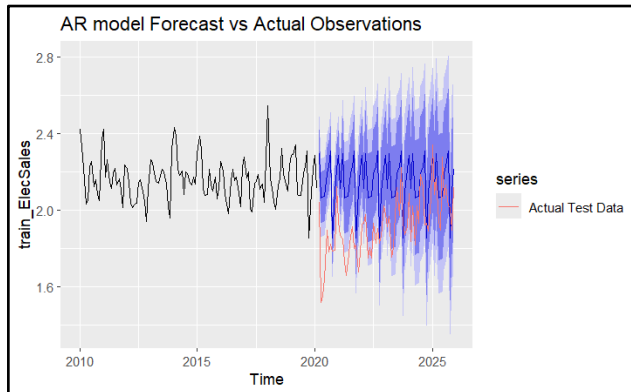
This R code performs Time Series Cross-Validation (tsCV) to compare the predictive accuracy of a custom ARIMA model against an ETS model. It calculates the Root Mean Squared Error (RMSE) for both, allowing you to determine which model produces more reliable 12-step ahead forecasts.

The time series cross-validation results indicate that the MA model performs better than the AR model, as it yields a lower forecasting error (0.09039 compared to 0.1066). This suggests that the MA model provides slightly more accurate out-of-sample predictions for the given dataset, although the difference in performance is relatively small.

5.6 Autolayer and Prediction

```
{r}
autoplot(fc_ma) +
  autolayer(test_ElecSales, series="Actual Test Data") + ggtitle("MA model Forecast vs Actual Observations")
```

→ **autoplot()**: Plots the forecasts from the MA-



The blue line and shaded regions represent the forecasted values and their prediction intervals, while the red line represents the actual observed values from the testing dataset. The MA model closely follows the stable yearly seasonal pattern, while the AR model shows more short-term fluctuations as it captures higher-frequency dynamics in the series. However, both models show large deviations during the 2020 COVID-19 period, where the sudden structural shock could not be captured using historical pattern alone.

During the disruption, both models continued to rely on past behaviour, with the AR model showing greater uncertainty through wider prediction intervals. As the series stabilises after 2020, both forecasts gradually align more closely with the actual values by 2024, indicating improved fit under normalised conditions. Therefore, the MA model provides a smoother and more stable representation of seasonality, while AR model better reflects short-term volatility in the data.

DATASET 3 : End-use energy consumption in the transportation sector (TNACBUS)

5.1 AR Model Estimation

```
> # Stage 5: Build Pure AR and MA Models ####
> ## 5.1 AR Model Estimation ###
> # Define training data
> y_train <- TNACBUS_train
> # AR Models: AR(p)
> ar1 <- Arima(y_train, order = c(1,1,0))
> ar2 <- Arima(y_train, order = c(2,1,0))
> ar3 <- Arima(y_train, order = c(3,1,0))
> # Compare AR models
> AIC(ar1, ar2, ar3)
      df      AIC
ar1  2 1475.475
ar2  3 1476.151
ar3  4 1476.504
```

→ **Arima()**: Estimates pure autoregressive (AR) models where future values depends on past observation

The Akaike Information Criterion (AIC) result clearly indicates that AR(1) is the most efficient autoregressive specification for TNACBUS energy consumption as it produces the lowest AIC value compared to AR(2) and AR(3). This suggests that current energy consumption is primarily influenced by the most recent previous observation, while additional lag terms contribute limited improvement to model fit. The result reflects strong short-term persistence in transportation energy demand, where recent consumption behaviour remains highly relevant in predicting future values.

5.2 MA Model Estimation

```
> ## 5.2 MA Model Estimation ####
> # MA Models: MA(q)
> ma1 <- Arima(y_train, order = c(0,1,1))
> ma2 <- Arima(y_train, order = c(0,1,2))
> ma3 <- Arima(y_train, order = c(0,1,3))
> # Compare MA models
> AIC(ma1, ma2, ma3)
      df      AIC
ma1    2 1478.691
ma2    3 1474.702
ma3    4 1475.535
```

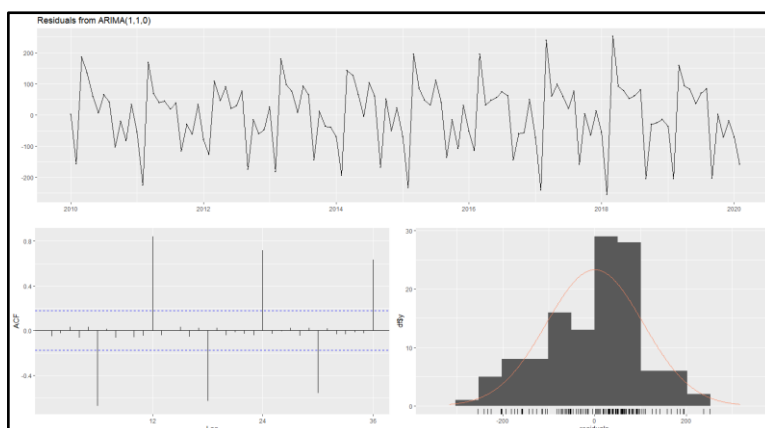
→ **Arima()**: Estimates pure moving average (MA) models where future values depends on past error terms

The AIC results for the MA models indicate that MA(1) generally provides the best or most efficient fit among the competing MA structures. This indicates that unexpected shocks affecting transportation energy consumption tend to have relatively short-lived effects. Most disturbances are absorbed within one period and do not persist for many months. This behaviour is consistent with the transportation sector, where temporary disruptions such as fuel price fluctuations, weather conditions, or short-term economic uncertainty may affect consumption briefly but do not create prolonged deviations from the overall trend.

5.3 Residual Diagnostics

```
# Residual diagnostics for selected AR and MA models
checkresiduals(ar1)
checkresiduals(ma1)
```

→ **checkresiduals()**: Tests whether residuals behave like white noise hv

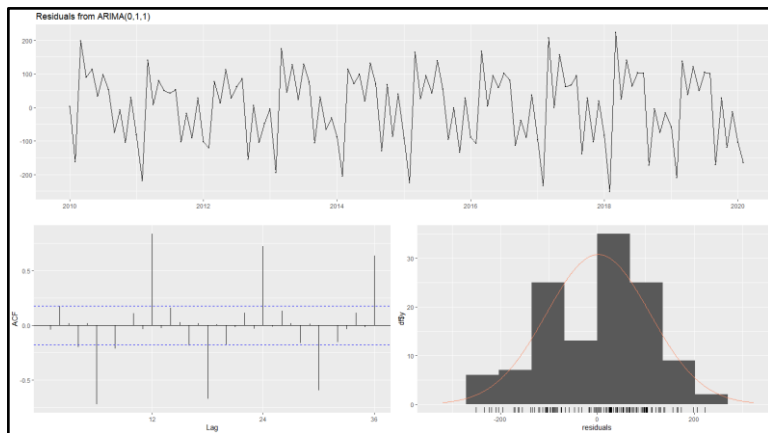


```
> checkresiduals(ar1)

Ljung-Box test

data: Residuals from ARIMA(1,1,0)
Q* = 298.87, df = 23, p-value < 2.2e-16

Model df: 1. Total lags used: 24
```



```
> checkresiduals(mal)

Ljung-Box test

data: Residuals from ARIMA(0,1,1)
Q* = 343.75, df = 23, p-value < 2.2e-16

Model df: 1. Total lags used: 24
```

The Ljung-Box test strongly rejects the null hypothesis of no autocorrelation in the residuals for both AR(1) and MA(1), indicating that the residuals are not white noise and that both models fail to fully capture the underlying time series structure.

Although AR(1) performs slightly better in terms of AIC, neither model removes serial dependence, showing that both are underfitting the data. This is expected in TNACBUS energy consumption due to strong seasonality, a structural break from COVID-19 and non-linear recovery patterns. Therefore, pure AR and MA models are too simplistic, justifying the need for more advanced models such as ARIMA and ETS in the next stage.

5.4 Forecasting Performance Evaluation

```
> ## 5.4 Forecasting Performance Evaluation ###
> # Generate forecasts using test set horizon
> fc_ar1 <- forecast(ar1, h = length(TNACBUS_test))
> fc_mal <- forecast(mal, h = length(TNACBUS_test))
> # Accuracy comparison against test set
> accuracy(fc_ar1, TNACBUS_test)
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1 Theil's U
Training set 0.8059443 105.2263  84.4418 -0.1137531 3.776974 2.196036 -0.05128724 NA
Test set    71.9743449 187.0534 158.4080  2.5167626 7.121257 4.119640  0.65532624 1.285837
> accuracy(fc_mal, TNACBUS_test)
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1 Theil's U
Training set 2.137472 106.6208  89.0325 -0.07846719 3.965730 2.315424 -0.03988985 NA
Test set    20.102978 173.8860 136.0602  0.21911799 6.274642 3.538450  0.64985055 1.238398
```

→ **forecast():** Generates forecasts over the test period

Both models perform better on the training data than on the test data, indicating reduced predictive accuracy when applied to unseen observations. The MA(1) model performs better than the AR(1) model because it produces lower RMSE, MAE and MAPE values during the test period. This suggests that transportation energy consumption is more strongly influenced by short-term shocks and irregular fluctuations than by long autoregressive persistence. However, both models still show relatively large forecasting errors during the pandemic and recovery period, while Theil's U values above 1 indicate that neither model outperforms a naive benchmark forecast.

5.5 Rolling Forecast Validation (tsCV)

```
> ## 5.5 Rolling Forecast Validation (tsCV) ####
> # Rolling forecast validation for AR(1)
> error_ar1 <- tscv(
+   y_train,
+   function(x, h)
+     forecast(Arima(x, order = c(1,1,0)), h = h),
+   h = 12
+ )
> # Rolling forecast validation for MA(1)
> error_ma1 <- tscv(
+   y_train,
+   function(x, h)
+     forecast(Arima(x, order = c(0,1,1)), h = h),
+   h = 12
+ )
> # RMSE comparison
> sqrt(mean(error_ar1^2, na.rm = TRUE))
[1] 129.4302
> sqrt(mean(error_ma1^2, na.rm = TRUE))
[1] 121.4933
```

→ **tsCV()**: Perform rolling forecast cross-validation, where the model is repeatedly re-estimated and tested across multiple forecast horizons (e.g., 12 months)

The rolling cross-validation results show that the MA(1) model provides more stable forecasting performance across time compared to the AR(1) model. This suggests that transportation energy consumption is more strongly influenced by short-term shocks and irregular demand fluctuations than by long-term autoregressive dependence, making MA models more suitable than AR models for this dataset.

5.6 AR & MA Model Forecast VS Actual

```
## 5.6 Forecast VS Actual ####
# AR(1) forecast
fc_ar1 <- forecast(ar1, h = length(TNACBUS_test))

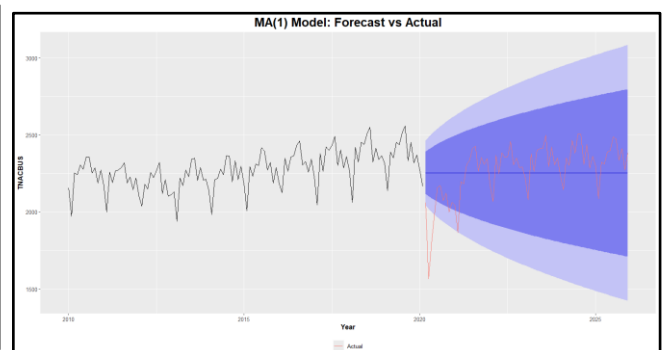
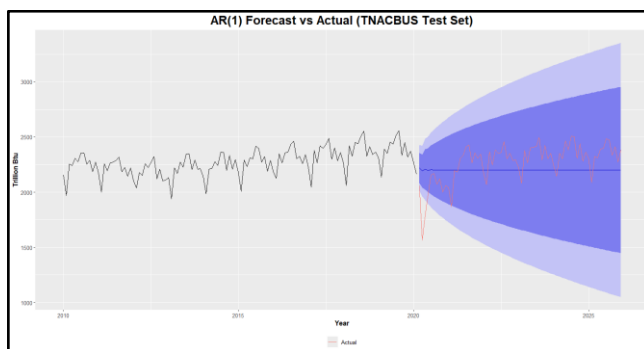
# Plot AR forecast vs actual
autoplot(fc_ar1) +
  autolayer(TNACBUS_test, series = "Actual") +
  ggtitle("AR(1) Forecast vs Actual (TNACBUS Test Set)") +
  xlab("Year") +
  ylab("Trillion Btu") +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5, size = 20),
    axis.title = element_text(face = "bold"),
    legend.title = element_blank(),
    legend.position = "bottom")

# MA(1) forecast
fc_ma1 <- forecast(ma1, h = length(TNACBUS_test))

# Plot MA forecast vs actual
autoplot(fc_ma1) +
  autolayer(TNACBUS_test, series = "Actual") +
  ggtitle("MA(1) Model: Forecast vs Actual") +
  xlab("Year") +
  ylab("TNACBUS") +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(hjust = 0.5, size = 20, face = "bold"),
    axis.title = element_text(face = "bold"),
    legend.title = element_blank(),
    legend.position = "bottom")
```

→ **forecast()**: Generate AR(1) and MA(1) over the test horizon

autolayer(): Add actual test data for



The AR(1) model produces a more responsive forecast that reacts strongly to recent past values. This results in higher short-term fluctuations and greater sensitivity to changes in the series. However, during the COVID-19 period, the model significantly overestimates actual

values as it continues projecting pre-shock dynamics. Once the series stabilises post-2020, the forecasts gradually align more closely with observed values, indicating better performance under stable conditions.

In contrast, the MA(1) model generates a smoother and more stable forecast by relying on past error terms rather than past values. It effectively captures the underlying seasonal structure of the series but, similar to the AR model, it does not adequately account for the abrupt structural break during the COVID-19 period, resulting in systematic overestimation during the shock phase. However, as the series stabilises after 2020, the forecasts progressively align more closely with the actual observations, indicating improved accuracy under normalised post-pandemic conditions.

Therefore, the AR model is more volatile and responsive, while the MA model is smoother and more stable. Both perform well in stable periods but struggle to capture sudden structural breaks such as the COVID-19 shock.

DATASET 4: Electrical System Energy Losses Proportioned to the Transportation Sector (LOACBUS)

5.1 AR-Based Model Estimation & Diagnostics

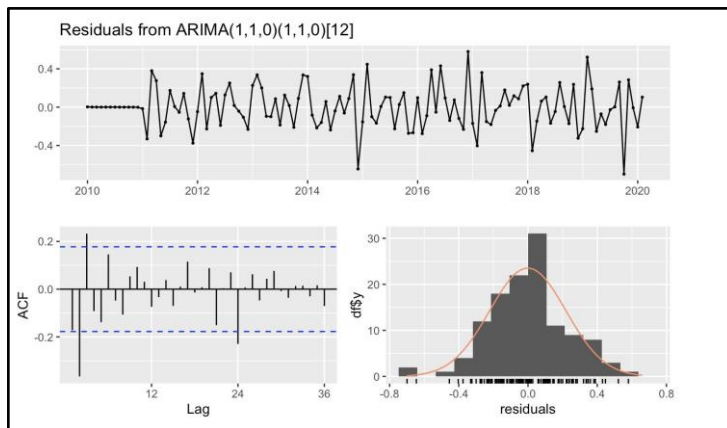
```
# 5.1 AR-BASED MODEL ESTIMATION & DIAGNOSTICS
# AR(1)
ar1 <- Arima(train_ElecSystemLoss,
             order=c(1,1,0),
             seasonal=c(1,1,0))

# AR(2)
ar2 <- Arima(train_ElecSystemLoss,
             order=c(2,1,0),
             seasonal=c(1,1,0))

# Compare AR models
AIC(ar1, ar2)

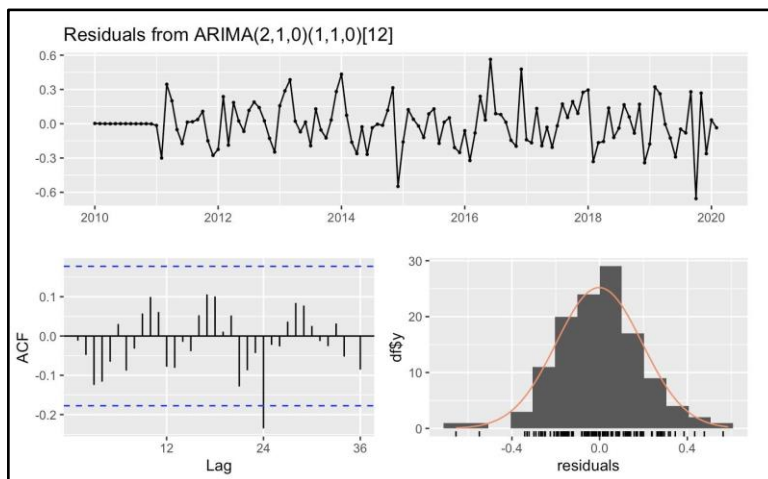
# Residual diagnostics
checkresiduals(ar1)
checkresiduals(ar2)
```

→ **Arima():** Estimate AR-based seasonal ARIMA models for electricity system losses based on the PACF behaviour identified in Stage 4.
→ **checkresiduals():** Examines residual



Ljung-Box test
data: Residuals from ARIMA(1,1,0)(1,1,0)[12]
 $Q^* = 54.457$, $df = 22$, $p\text{-value} = 0.0001418$
Model df: 2. Total lags used: 24

Based on the PACF cutoff near lag 2 and remaining seasonal spikes at lag 12, ARIMA(1,1,0)(1,1,0)[12] and ARIMA(2,1,0)(1,1,0)[12] were estimated as AR-based candidate models. The AR(1) model failed the Ljung-Box test with $p\text{-value} = 0.0001418$, indicating significant remaining autocorrelation and insufficient model adequacy. In addition, the model produced a relatively higher AIC value (-0.46), suggesting weaker overall goodness-of-fit. Residual ACF plots also showed several spikes outside the confidence bounds, suggesting that the model did not fully capture the seasonal behaviour in electricity system losses.



Ljung-Box test
data: Residuals from ARIMA(2,1,0)(1,1,0)[12]
 $Q^* = 26.518$, $df = 21$, $p\text{-value} = 0.1874$
Model df: 3. Total lags used: 24

Next, the AR(2) model showed substantial improvement, achieving a much lower AIC value (-23.95) together with a Ljung-Box $p\text{-value}$ of 0.1874. The residuals fluctuated randomly around zero, most ACF spikes remained within the confidence bounds, and the histogram was approximately bell-shaped and centred near zero. Therefore, ARIMA(2,1,0)(1,1,0)[12] was retained as the final AR-based model.

5.2 MA-Based Model Estimation & Diagnostics

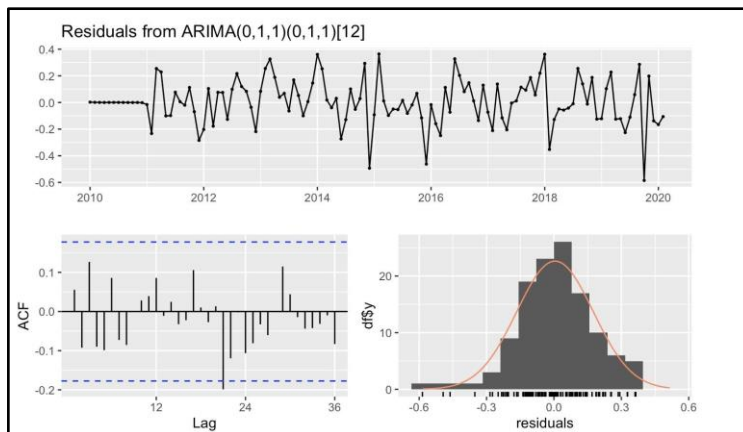
```
# 5.2 MA-BASED MODEL ESTIMATION & DIAGNOSTICS
# MA(1)
ma1 <- Arima(train_ElecSystemLoss,
              order=c(0,1,1),
              seasonal=c(0,1,1))

# MA(2)
ma2 <- Arima(train_ElecSystemLoss,
              order=c(0,1,2),
              seasonal=c(0,1,1))

# Compare MA models
AIC(ma1, ma2)

# Residual diagnostics
checkresiduals(ma1)
checkresiduals(ma2)
```

→ **Arima()**: Estimate MA-based models
 → **checkresiduals()**: Assess residual independence and remaining

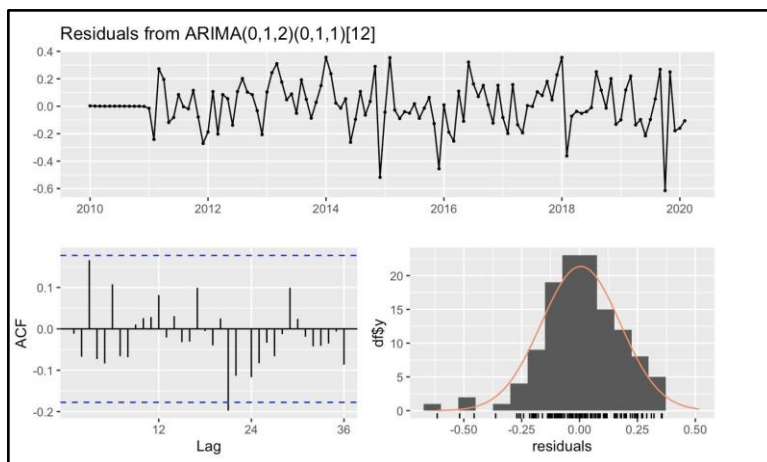


Ljung-Box test

data: Residuals from ARIMA(0,1,1)(0,1,1)[12]
 $Q^* = 21.679$, $df = 22$, $p\text{-value} = 0.4792$

Model df : 2. Total lags used: 24

The ARIMA(0,1,1)(0,1,1)[12] model produced residuals that fluctuated randomly around zero with most ACF spikes within the confidence bounds. The model also achieved the lowest AIC value (-40.13) among all AR and MA candidates, indicating the strongest overall goodness-of-fit. Furthermore, the Ljung-Box p-value of 0.4792 indicates that the residuals behave similarly to white noise, suggesting acceptable model adequacy.



Ljung-Box test

data: Residuals from ARIMA(0,1,2)(0,1,1)[12]
 $Q^* = 21.66$, $df = 21$, $p\text{-value} = 0.4193$

Model df : 3. Total lags used: 24

The ARIMA(0,1,2)(0,1,1)[12] model also achieved acceptable residual diagnostics with a Ljung-Box p-value above 0.05. Although the model produced slightly lower MAE and

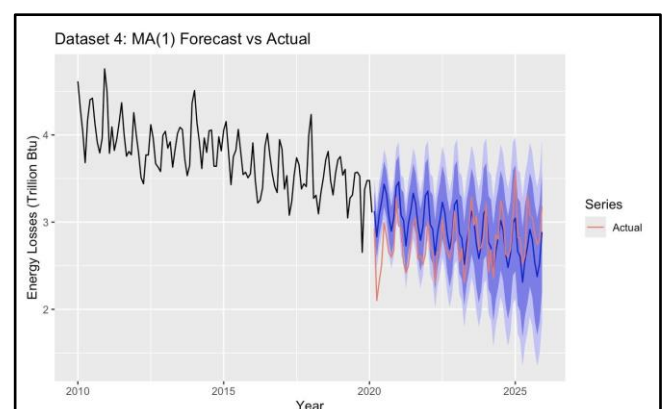
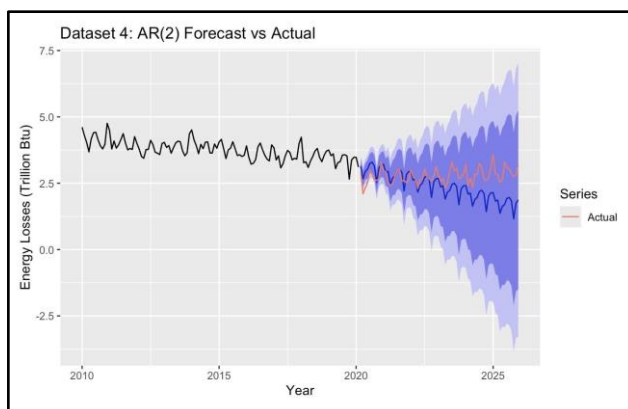
MAPE values, its AIC value (-38.88) was higher than the MA(1) model, while the improvement in forecasting accuracy remained minimal despite the additional parameter. Therefore, MA(1) was retained as the final MA-based model due to its better balance between forecasting accuracy, residual adequacy, and model simplicity.

5.3 Forecast Evaluation

```
# Forecasts
fc_ar2 <- forecast(ar2, h=h)
fc_ma1 <- forecast(ma1, h=h)

# MA(1) Forecast Plot
autoplot(fc_ma1) +
  autolayer(test_ElecSystemLoss,
    series="Actual") +
```

→ **forecast()**: Generate forecasts over the test period, → **autoplot()** and **autolayer()**: Compare forecasts against ..



The AR(2) and MA(1) models both captured the general seasonal movement in electricity system losses during the post-COVID recovery period. However, the MA(1) model followed the actual observations more closely and produced more stable forecasts, while the AR(2) model gradually diverged from the observed recovery pattern over time. Although most actual observations remained within the forecast intervals for both models, the AR(2) model generated substantially wider prediction intervals, indicating greater long-term forecasting uncertainty and weaker forecasting stability. In comparison, the MA(1) model maintained a more consistent seasonal structure and demonstrated stronger overall forecasting reliability under post-pandemic transportation and mobility conditions.

5.4 Traditional Accuracy

```
# 5.4 TRADITIONAL ACCURACY MEASURES
> accuracy(fc_ar2,
+ test_ElecSystemLoss)
+
+      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1 Theil's U
Training set -0.001963784 0.1953546 0.1454121 -0.2060761 3.96860 0.7466081 -0.01174227 NA
Test set    0.389475095 0.6847627 0.5440620 13.2418746 19.31199 2.7934476 0.79811211 2.133942
>
> accuracy(fc_ma1,
+ test_ElecSystemLoss)
+
+      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1 Theil's U
Training set 0.004761959 0.1696531 0.1264098 -0.03863687 3.453185 0.649042 0.05555221 NA
Test set   -0.12722616 0.3242562 0.2693310 -5.22881378 9.936608 1.382861 0.70928596 1.064662
```

→ **accuracy()**: Evaluate forecasting performance using RMSE MAE MAPE and Theil's U

The MA(1) model achieved lower RMSE (0.3243), MAE (0.2693), and MAPE (9.94%) values compared with the AR(2) model, indicating stronger forecasting accuracy during the COVID-19 and post-pandemic recovery period. The lower MAPE and Theil's U values further suggest that the MA(1) model handled short-term fluctuations and pandemic-related disruptions more effectively, producing forecasts closer to the actual observations overall.

5.5 Rolling Forecast Validation (tsCV)

```
> # Run tsCV
> cv_ar2 <- tsCV(train_ElecSystemLoss,
+               f_ar2,
+               h=12)
> cv_ma1 <- tsCV(train_ElecSystemLoss,
+               f_ma1,
+               h=12)
>
> # tsCV RMSE
> sqrt(mean(cv_ar2^2, na.rm=TRUE))
[1] 0.2686533
> sqrt(mean(cv_ma1^2, na.rm=TRUE))
[1] 0.2349001
```

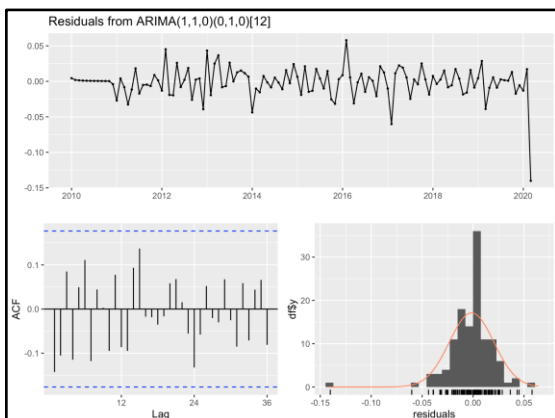
→ **tsCV()**: Using a 12-step forecasting horizon to evaluate rolling forecast

The MA(1) model achieved a lower tsCV RMSE value (0.2349) compared with the AR(2) model (0.2687), indicating stronger forecasting stability and better predictive performance across multiple forecast origins. This suggests that the MA(1) model captured short-term fluctuations in electricity system losses more effectively during the COVID-19 recovery period.

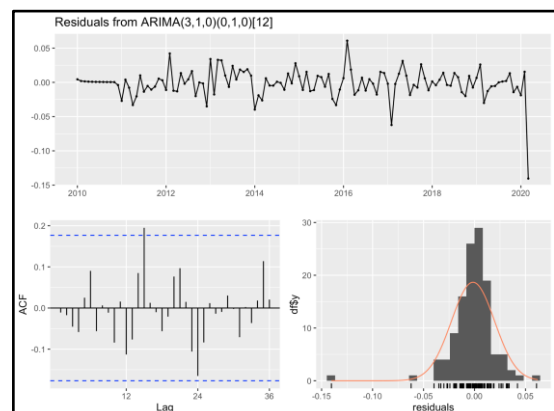
DATASET 5: Total Energy Consumed by The Transportation Sector (TEACBUS)

5.1 Pure AR Models & Reidentification

```
# Stage 5.1 Initial pure AR benchmark
fit_ar_initial <- Arima(train_ts,
+                      order = c(1, 1, 0),
+                      seasonal = c(0, 1, 0),
+                      lambda = lambda_used,
+                      biasadj = TRUE)
summary(fit_ar_initial)
checkresiduals(fit_ar_initial)
# Output file: figures/stage5_residual_AR1_initial_TotalEnergy.png
```



```
# Stage 5.2 Reidentified pure AR benchmark
fit_ar_reid <- Arima(train_ts,
+                    order = c(3, 1, 0),
+                    seasonal = c(0, 1, 0),
+                    lambda = lambda_used,
+                    biasadj = TRUE)
summary(fit_ar_reid)
checkresiduals(fit_ar_reid)
# Output files: residual diagnostics + separate pure AR forecast plot
```

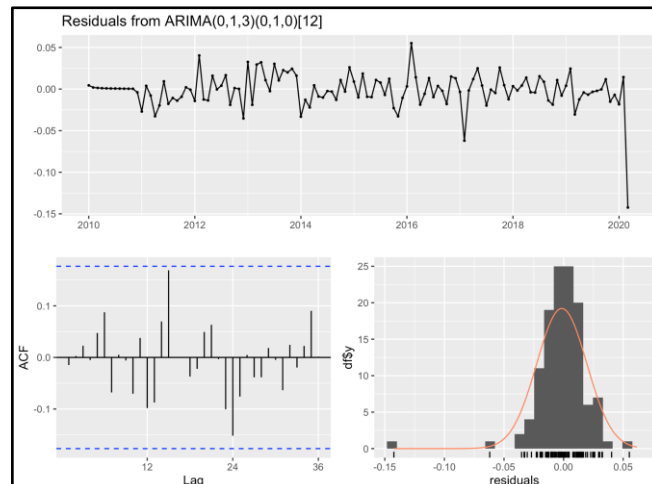
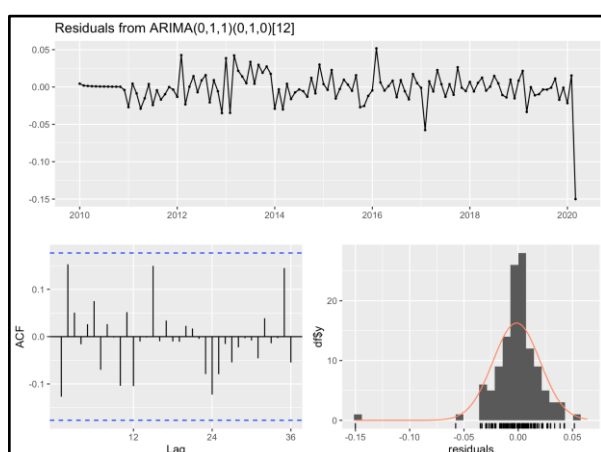


The initial pure AR benchmark is ARIMA(1, 1, 0)(0, 1, 0)₁₂ on the log scale. Its residual time plot is centred near zero, although the COVID boundary produces a large negative residual at the end of the training period. The residual ACF is mostly inside the significance bands, and the Ljung-Box p-value is 0.195. Therefore, the no-autocorrelation null is not rejected at the 5% level. Reidentification with ARIMA(3, 1, 0)(0, 1, 0)₁₂ tests whether adding more short-run AR terms improves the model. The AR(3) residuals are still centred near zero, but its Ljung-Box p-value is 0.062, which is acceptable but close to 0.05. The AR(3) model improves test RMSE relative to AR(1), but its residual adequacy is more borderline and its tsCV RMSE is worse than the best pure MA candidate.

5.2 Pure MA Models & Reidentification

```
# Stage 5.3 Initial pure MA benchmark
fit_ma_initial <- Arima(train_ts,
  order = c(0, 1, 1),
  seasonal = c(0, 1, 0),
  lambda = lambda_used,
  biasadj = TRUE)
summary(fit_ma_initial)
checkresiduals(fit_ma_initial)
# This model is retained because it passes the Ljung-Box check and gives the best Stage 5 forecast accuracy.
```

```
# Stage 5.4 Reidentified pure MA benchmark tested for completeness
fit_ma_reid <- Arima(train_ts,
  order = c(0, 1, 3),
  seasonal = c(0, 1, 0),
  lambda = lambda_used,
  biasadj = TRUE)
summary(fit_ma_reid)
checkresiduals(fit_ma_reid)
# Reidentification is reported, but the comparison table decides by residual validity and tsCV accuracy.
```

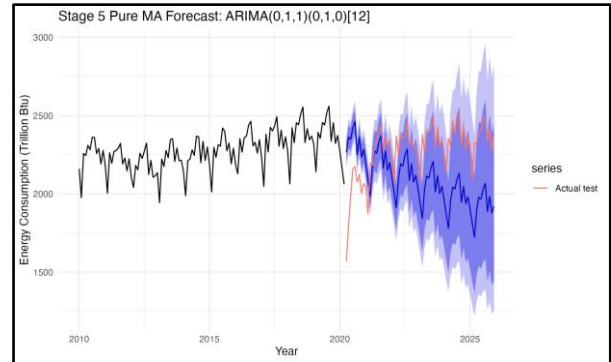
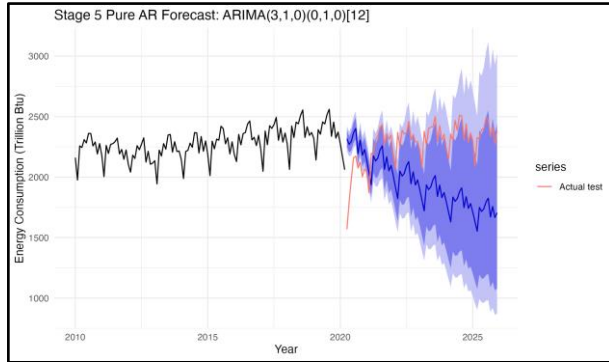


The initial pure MA benchmark is ARIMA(0, 1, 1)(0, 1, 0)₁₂. Its residual diagnostics show no strong remaining autocorrelation: most ACF bars remain within the confidence limits, and the Ljung-Box p-value is 0.163. The histogram is concentrated near zero, with one large tail observation associated with the COVID shock. The reidentified ARIMA(0, 1, 3)(0, 1, 0)₁₂ also passes the residual screen with p-value 0.102, but the additional MA terms do not improve the main validation metric. Because MA(1) is already residual-acceptable and has stronger tsCV and test accuracy, MA(3) is reported as a reidentified candidate rather than selected. This is a useful example of why reidentification should improve forecast validation, not merely add parameters.

5.3 Stage 5 Comparison Table and Separate Forecast Plots

Model	Ljung-Box p	Residual OK?	tsCV RMSE	tsCV MAE	Test RMSE	Test MAE	Test MAPE	AICc
ARIMA(0,1,1)(0,1,0) ₁₂	0.163	Yes	38.637	39.135	331.601	293.449	12.994	-504.321
ARIMA(2,1,0)(0,1,0) ₁₂	0.086	Yes	44.524	44.635	502.397	450.722	19.633	-508.222
ARIMA(1,1,0)(0,1,0) ₁₂	0.195	Yes	46.967	47.108	605.593	541.980	23.476	-508.805
ARIMA(3,1,0)(0,1,0) ₁₂	0.062	Yes	47.515	45.859	481.707	431.997	18.845	-502.880
ARIMA(0,1,3)(0,1,0) ₁₂	0.102	Yes	48.240	41.415	349.006	310.313	13.709	-499.154
ARIMA(0,1,2)(0,1,0) ₁₂	0.130	Yes	59.110	46.186	451.963	405.427	17.729	-503.605

Stage 5 Pure AR Forecast: ARIMA(3,1,0)(0,1,0)[12] Stage 5 Pure MA Forecast: ARIMA(0,1,1)(0,1,0)[12]



The comparison table is ordered by the required decision logic: first the Ljung-Box residual screen, then tsCV RMSE/MAE, then test accuracy, and finally AICc. All six Stage 5 candidates pass the residual screen, so the next criterion is rolling-origin validation. MA(1) has the lowest tsCV RMSE and also the lowest test RMSE, MAE, and MAPE. AICc alone favours AR(1) or AR(2), but AICc is in-sample goodness-of-fit evidence and should not override residual adequacy plus validation accuracy. The selected Stage 5 pure benchmark is therefore ARIMA(0, 1, 1)(0, 1, 0)₁₂. Both pure AR and pure MA forecasts miss the initial COVID collapse because a pre-pandemic seasonal model cannot anticipate an external mobility shock, but MA(1) gives the stronger overall forecast comparison.

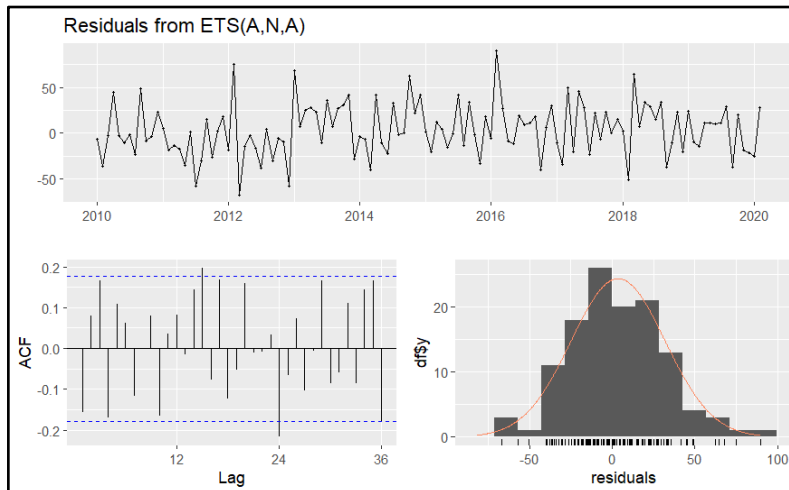
Stage 6: Auto ETS and Auto ARIMA

Dataset 1: Total Primary Energy Consumed by the Transportation Sector (TXABUS)

6.1.1 Auto ETS

```
fit_ets_Trans <- ets(train_TotalPrimaryEnergy_Trans)
summary(fit_ets_Trans)
checkresiduals(fit_ets_Trans)
```

→ **ets()**: Automatically selects the best



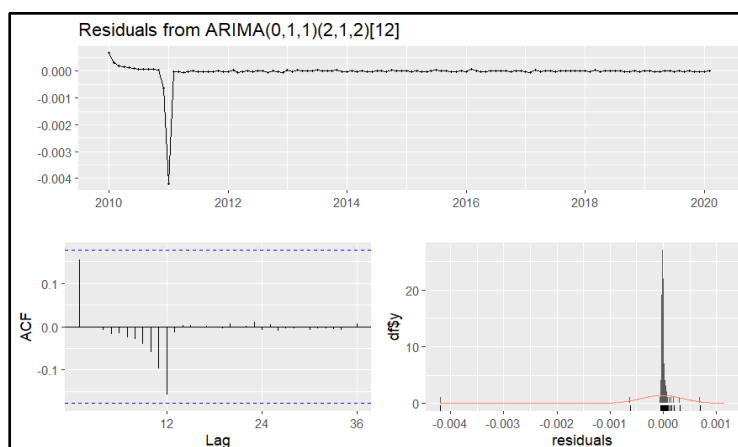
Ljung-Box test
data: Residuals from ETS(A,N,A)
 $Q^* = 47.689$, $df = 24$, $p\text{-value} = 0.002757$
Model df: 0. Total lags used: 24

The residuals fluctuate randomly around zero but show spikes around 2016 and 2020 due to the COVID-19 structural break. The ACF reveals several significant spikes and the histogram displays a slight right skew, indicating remaining autocorrelation and that the model does not fully capture the dependence structure of the series. This is confirmed by the Ljung-Box test ($Q^* = 47.689$, $p\text{-value} = 0.002757$), which rejects the white-noise hypothesis, suggesting that the ETS(A,N,A) model alone is insufficient to model this series.

6.1.2 Auto ARIMA

```
fit_aa_Trans <- auto.arima(train_TotalPrimaryEnergy_Trans,
                           lambda = "auto",
                           stepwise = FALSE,
                           approximation = FALSE)
summary(fit_aa_Trans)
checkresiduals(fit_aa_Trans)
```

→ **auto.arima()**: Automatically selects the best-fitting ARIMA model using full

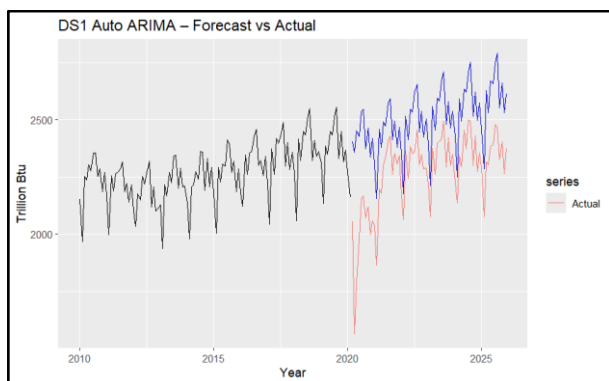


Ljung-Box test
data: Residuals from ARIMA(0,1,1)(2,1,2)[12]
 $Q^* = 8.8151$, $df = 19$, $p\text{-value} = 0.9764$
Model df: 5. Total lags used: 24

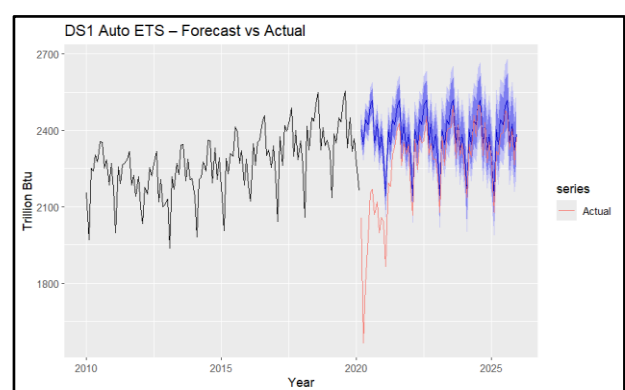
The residuals show a large spike around 2010–2011 followed by near-zero values, suggesting numerical issues from the Box-Cox transformation ($\lambda \approx 2$). The ACF shows significant spikes at lag 12 and the histogram is abnormally narrow and concentrated around zero, indicating possible overfitting. Despite this, the Ljung-Box test ($Q^* = 8.8151$, $p\text{-value} = 0.9764$) fails to reject the null hypothesis, confirming that the residuals are statistically white noise and that the model has captured the underlying patterns in the series, though the visual diagnostics suggest its reliability should be interpreted with caution.

6.1.3 Both Models' Forecast Plot

```
fc_ets_Trans <- forecast(fit_ets_Trans, h = h_Trans)
autoplot(fc_ets_Trans) +
  autolayer(test_TotalPrimaryEnergy_Trans, series = "Actual") +
  ggtitle("DS1 Auto ETS - Forecast vs Actual") +
  xlab("Year") + ylab("Trillion Btu")
```



```
fc_aa_Trans <- forecast(fit_aa_Trans, h = h_Trans)
autoplot(fc_aa_Trans) +
  autolayer(test_TotalPrimaryEnergy_Trans, series = "Actual") +
  ggtitle("DS1 Auto ARIMA - Forecast vs Actual") +
  xlab("Year") + ylab("Trillion Btu")
```



Both the Auto ETS and Auto ARIMA models capture the main seasonal pattern, but neither reflects the structural change caused by the COVID-19 pandemic. Since both models were trained on data from before the pandemic, they miss the sharp drop in demand in early 2020. The Auto ETS model is more reliable, staying closer to actual values and giving narrower prediction intervals of about 2,000 to 2,700 trillion Btu by 2025 to 2026. However, actual values are usually at the lower end of these intervals, which suggests the model slightly overestimates post-pandemic consumption. The Auto ARIMA model, on the other hand, does not show any visible prediction intervals, which is unusual and reflects the overfitting concern from the Box-Cox transformation (λ about 2). Its point forecast also overestimates actual values by around 200 to 300 trillion Btu during the test period and continues to rise, missing the stabilization in post-pandemic consumption. Overall, the Auto ARIMA model is not reliable for this dataset. The Auto ETS or a manually tuned ARIMA model would be better choices.

6.1.4 Both Models' Traditional and tsCV Accuracy

```
acc_ets_Trans <- accuracy(fc_ets_Trans$mean, test_TotalPrimaryEnergy_Trans)
cat("ETS Traditional Accuracy:\n")
print(acc_ets_Trans)
fets <- function(x, h) forecast(ets(x), h = h)
cv_ets <- tsCV(train_TotalPrimaryEnergy_Trans, fets, h = 12)
cv_ets_rmse <- sqrt(mean(cv_ets^2, na.rm = TRUE))
cv_ets_mae <- mean(abs(cv_ets), na.rm = TRUE)
cat(sprintf("Auto ETS | tsCV RMSE: %.4f | tsCV MAE: %.4f\n",
            cv_ets_rmse, cv_ets_mae))
```

```
acc_aa_Trans <- accuracy(fc_aa_Trans$mean, test_TotalPrimaryEnergy_Trans)
cat("Auto ARIMA Traditional Accuracy:\n")
print(acc_aa_Trans)
aa_order <- fit_aa_Trans$arma
p_aa <- aa_order[1]; q_aa <- aa_order[2]
p_aa <- aa_order[3]; q_aa <- aa_order[4]
d_aa <- aa_order[6]; q_aa <- aa_order[7]
faa_fixed <- function(x, h) {
  forecast(Arima(x,
    order = c(p_aa, d_aa, q_aa),
    seasonal = list(order = c(p_aa, d_aa, q_aa), period = 12),
    lambda = fit_aa_Trans$lambda),
    h = h)
}
cv_aa <- tsCV(train_TotalPrimaryEnergy_Trans, faa_fixed, h = 12)
cv_aa_rmse <- sqrt(mean(cv_aa^2, na.rm = TRUE))
cv_aa_mae <- mean(abs(cv_aa), na.rm = TRUE)
cat(sprintf("Auto ARIMA | tsCV RMSE: %.4f | tsCV MAE: %.4f\n",
            cv_aa_rmse, cv_aa_mae))
```

```
ETS Traditional Accuracy:
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set -113.905 185.2188 114.5377 -5.550011 5.57577 0.852529 1.399181
Auto ETS | tsCV RMSE: 62.1735 | tsCV MAE: 44.4032
```

```
Auto ARIMA Traditional Accuracy:
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set -234.7233 259.8209 234.7233 -10.71608 10.71608 0.7450219 1.834217
Auto ARIMA | tsCV RMSE: 40.6464 | tsCV MAE: 32.3983
```

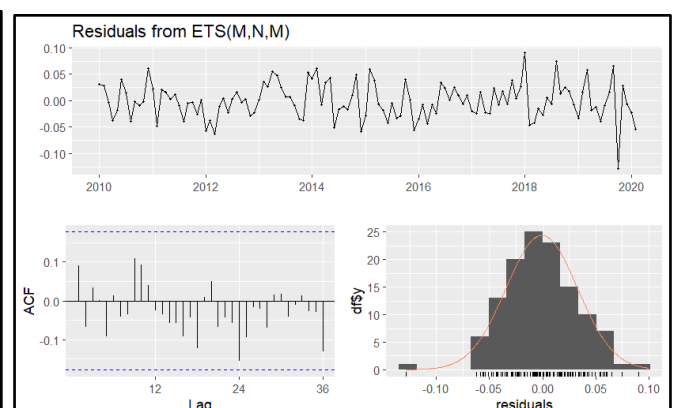
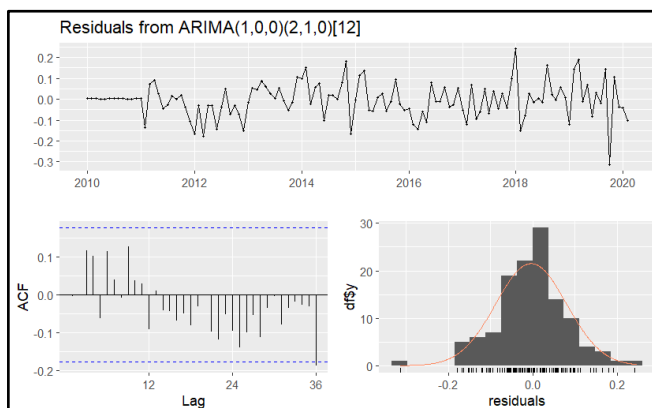
Both models show consistent overestimation (negative ME) with Theil's U above 1, indicating worse performance than a naïve forecast. Auto ETS outperforms Auto ARIMA across all test set metrics with lower RMSE (185.22 vs 259.82) and MAE (114.54 vs 234.72). Auto ARIMA's higher tsCV RMSE is on the original scale and combined with its abnormal residual diagnostics, confirms it should be discarded in favour of Auto ETS.

Dataset 2: Electricity Sales to Ultimate Customers in the Transportation Sector (ESACBUS)

6.2.1 Auto ETS & Auto ARIMA

```
{r}
# Fit the auto arima model
fit_auto_arima <- auto.arima(train_ElecSales, stepwise = FALSE, approximation =
FALSE)
summary(fit_auto_arima)

# Fit the ETS model
fit_auto_ets <- ets(train_ElecSales)
summary(fit_auto_ets)
```



The `auto.arima()` function is used to automatically identify the most appropriate ARIMA model for the training dataset by selecting the combination of parameters that minimises information criteria such as AIC, AICc, and BIC. Setting `stepwise = FALSE` and `approximation = FALSE` allows the function to perform a more exhaustive search across multiple model combinations, improving the reliability of the selected model. The `ets()` function is then used to fit an Exponential Smoothing State Space (ETS) model to the same dataset. Finally, the `summary()` function is applied to both models to display the selected model structures, estimated parameters, and training accuracy measures.

Auto Arima Model:

The `auto.arima()` function selected the $ARIMA(1,0,0)(2,1,0)[12]$ model for the dataset. This model includes: non-seasonal autoregressive component of order 1, seasonal differencing and two seasonal autoregressive terms operating at a 12-month seasonal period. The training accuracy measures show an RMSE value of 0.0836 and a MAPE value of 2.83%, suggesting that the model was able to fit the historical training data effectively while maintaining relatively low forecasting errors.

ETS Model:

The ETS selected has the structure $ES(M,N,M)$. This indicates there are multiplicative errors, no trend component and multiplicative seasonality. The absence of a trend component suggests that the dataset does not exhibit a strong long-term upward or downward movement, while the multiplicative seasonal component indicates that the seasonal fluctuations vary proportionally with the level of the series. The training accuracy measures produced an RMSE value of 0.0734 and a MAPE value of 2.63%, indicating that the ETS model was able to capture the historical seasonal patterns effectively with relatively low forecasting error.

Both the models have non-significant Ljung-box test results ($p > 0.05$), indicating that the residuals are independently distributed and resemble white noise. This basically means that both the models are adequately capturing the autocorrelation structure in the data. This is further supported by the ACF plot for both the models.

6.2.2 Both Models' Forecast Plot

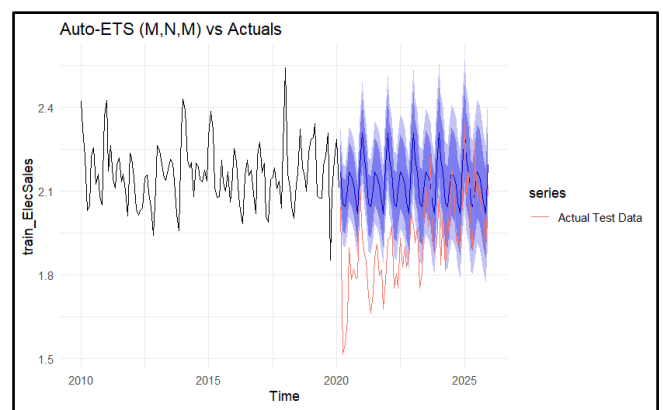
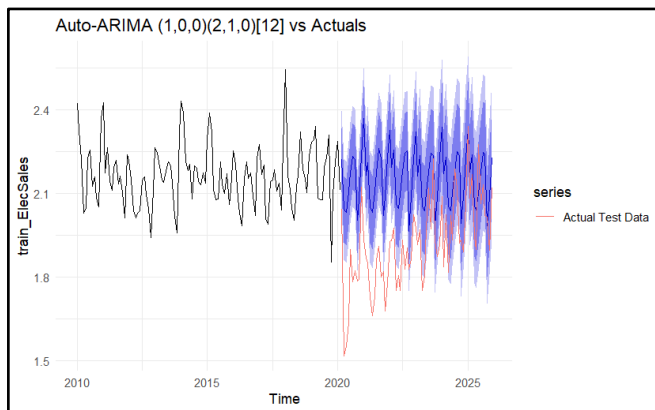
```
{r}
# Generate forecasts from the automated models
fc_auto_arima <- forecast(fit_auto_arima, h = length(test_ElecSales))
fc_auto_ets <- forecast(fit_auto_ets, h = length(test_ElecSales))

# Plot 1: Auto-ARIMA vs Actuals
p1 <- autoplot(fc_auto_arima) +
  autolayer(test_ElecSales, series="Actual Test Data") +
  ggtitle("Auto-ARIMA (1,0,0)(2,1,0)[12] vs Actuals") +
  theme_minimal()

# Plot 2: Auto-ETS vs Actuals
p2 <- autoplot(fc_auto_ets) +
  autolayer(test_ElecSales, series="Actual Test Data") +
  ggtitle("Auto-ETS (M,N,M) vs Actuals") +
  theme_minimal()

# Display them
print(p1)
print(p2)
```

```
{r}
# Checking the residuals
checkresiduals(fit_auto_arima)
checkresiduals(fit_auto_ets)
```



The ARIMA model forecasts electricity sold to transportation to remain stable, but completely fails to predict the sharp COVID-19 drop in 2020 where actuals crash well below the forecast range. However, by the end of the forecast horizon, the red actual line gradually converges back toward the blue forecast, suggesting the model recovers its accuracy in later years.

The ETS model tells a very similar story, equally failing to capture the 2020 COVID-19 crash. The key subtle difference is that the ETS forecast appears slightly smoother and less volatile in its seasonal swings compared to ARIMA, with somewhat narrower confidence bands reflecting its multiplicative error structure. Despite this minor difference, both models perform comparably poorly during the pandemic shock, and both see actuals gradually recover toward the forecast range toward the end.

6.2.4 Both Models' Traditional and tsCV Accuracy

```
{r}
# Check accuracy against the test set
accuracy(fc_auto_arma, test_ElecSales)
accuracy(fc_auto_ets, test_ElecSales)
```

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	-0.003871318	0.08356822	0.06113418	-0.2820038	2.82646	0.7821307	-0.004185113	NA
Test set	-0.237217437	0.27574437	0.24371617	-12.9338857	13.22474	3.1180249	0.462762296	1.782515

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	-0.001736041	0.07344673	0.05711906	-0.186238	2.630525	0.7307625	0.09689977	NA
Test set	-0.212383209	0.25355139	0.22154754	-11.674232	12.079642	2.8344068	0.53336011	1.650947

Both models perform well on the training set, with low error values, but deteriorate significantly on the test set largely due to the COVID-19 shock in 2020 that neither model could anticipate. Comparing the two, the ETS model slightly outperforms ARIMA across all key test set metrics : lower RMSE (0.254 vs 0.276), lower MAE (0.222 vs 0.244) and lower MAPE (12.1% vs 13.2%).

```
{r}
f_auto_arma_final <- function(x, h) {forecast(Arima(x, order=c(1,0,0), seasonal
=c(2,1,0)), h=h)}

f_auto_ets_final <- function(x, h) {forecast(ets(x), h=h)}

# Running the tsCV
cv_auto_arma <- tsCV(train_ElecSales, f_auto_arma_final, h=12)
cv_auto_ets <- tsCV(train_ElecSales, f_auto_ets_final, h=12)

# 3. Calculate RMSE
rmse_cv_auto_arma <- sqrt(mean(cv_auto_arma^2, na.rm=TRUE))
rmse_cv_auto_ets <- sqrt(mean(cv_auto_ets^2, na.rm=TRUE))

print(rmse_cv_auto_arma)
print(rmse_cv_auto_ets)
```

```
[1] 0.09556642
[1] 0.1646117
```

The code creates two forecasting functions; one for ARIMA and one for ETS. They are conducted using rolling cross-validation on the training data, forecasting 12 months ahead each time. It then calculates RMSE to see which model performs better. ARIMA comes out on top here with a lower RMSE (0.096) compared to ETS (0.165), meaning it does a better job forecasting electricity sold to transportation under normal conditions. To test this, both models were practiced on different parts of the training data multiple times, and ARIMA consistently did better each time. However, when both models were tested on real unseen data, they both struggled due to the COVID-19 drop in 2020, which makes sense since neither model had ever seen anything like a pandemic before.

Dataset 3: End-Use Energy Consumed by the Transportation Sector (TNACBUS)

6.1 Auto ETS & Auto ARIMA

```
> ## 6.1 ETS Model Estimation ###
> ets_model <- ets(y_train)
> summary(ets_model)
ETS(A,N,A)

call:
ets(y = y_train)

Smoothing parameters:
alpha = 0.3153
gamma = 1e-04

Initial states:
l = 2242.9046
s = 2.6481 -55.3586 46.7908 -28.0589 139.6601 117.0898
    44.4213 64.5011 -35.3593 23.8587 -235.6334 -84.5596

sigma: 30.3739

      AIC      AICC      BIC
1434.134 1438.663 1476.195

Training set error measures:
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 3.606762 28.57804 22.56204 0.1432726 0.9982743 0.5867599 -0.157973
```

```
> ## 6.3 Auto ARIMA Model Estimation ###
> arima_model <- auto.arima(y_train)
> summary(arima_model)
Series: y_train
ARIMA(2,1,0)(0,1,1)[12]

Coefficients:
      ar1      ar2      sma1
    -0.8153   -0.4038   -0.8472
s.e.    0.0876    0.0877    0.1371

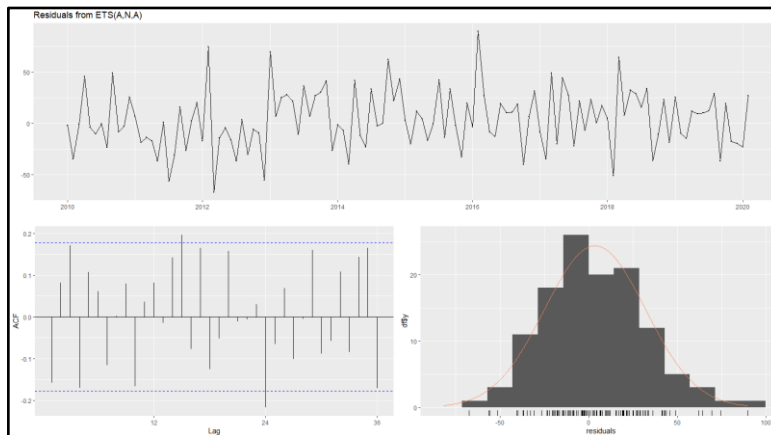
sigma^2 = 964.2; log likelihood = -535.36
AIC=1078.71  AICC=1079.1  BIC=1089.48

Training set error measures:
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 1.14451 28.94456 21.92283 0.03340371 0.9735415 0.5701361 -0.00381534
```

The `ets()` function is used to automatically fit an Exponential Smoothing State Space Model to the training data. The selected ETS(A,N,A) model indicates additive error, no trend, and additive seasonality, showing that the TNACBUS series mainly contains stable seasonal patterns after transformation. The smoothing parameter alpha of 0.3153 suggests that the model gives moderate weight to recent observations, while the near-zero gamma value indicates highly stable seasonality over time. In addition, the low RMSE and MAE values show that the ETS model fits the training data well and captures the historical seasonal movements effectively.

The selected ARIMA model incorporates both non-seasonal and seasonal differencing, confirming that the transportation energy consumption series contains strong trend and seasonal patterns that must be removed to achieve stationarity. The seasonal MA component indicates that seasonal shocks from previous periods continue to influence current energy consumption levels. In addition, the model demonstrates strong in-sample accuracy, suggesting that it captures the historical dynamics and seasonal structure of the series effectively during the training period.

6.2 ETS model & ARIMA model residual diagnostics

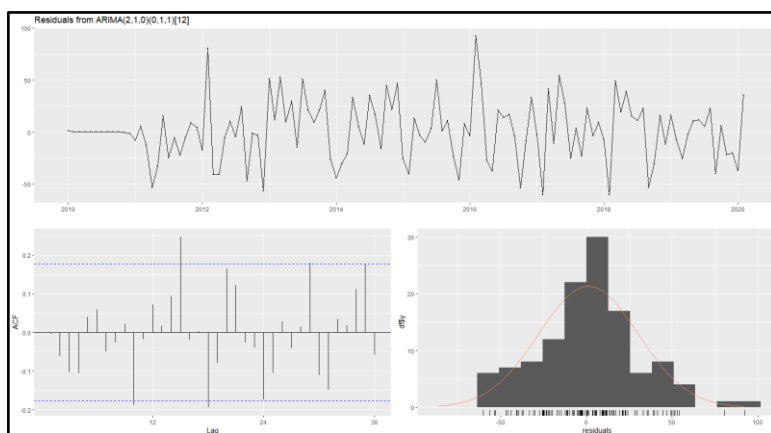


```
> checkresiduals(ets_model)

Ljung-Box test

data: Residuals from ETS(A,N,A)
Q* = 47.704, df = 24, p-value = 0.002746

Model df: 0.    Total lags used: 24
```



```
> checkresiduals(arima_model)

Ljung-Box test

data: Residuals from ARIMA(2,1,0)(0,1,1)[12]
Q* = 37.504, df = 21, p-value = 0.01473

Model df: 3.    Total lags used: 24
```

Both the ETS and ARIMA models show significant residual autocorrelation as both p-values are below 0.05. This indicates that neither model fully captures the underlying structure of TNACBUS energy consumption, and the residuals are not pure white noise. Although ARIMA performs slightly better than ETS, both still leave systematic patterns unexplained. This is consistent with earlier results showing reduced accuracy during structural changes such as COVID-19 and post-pandemic recovery.

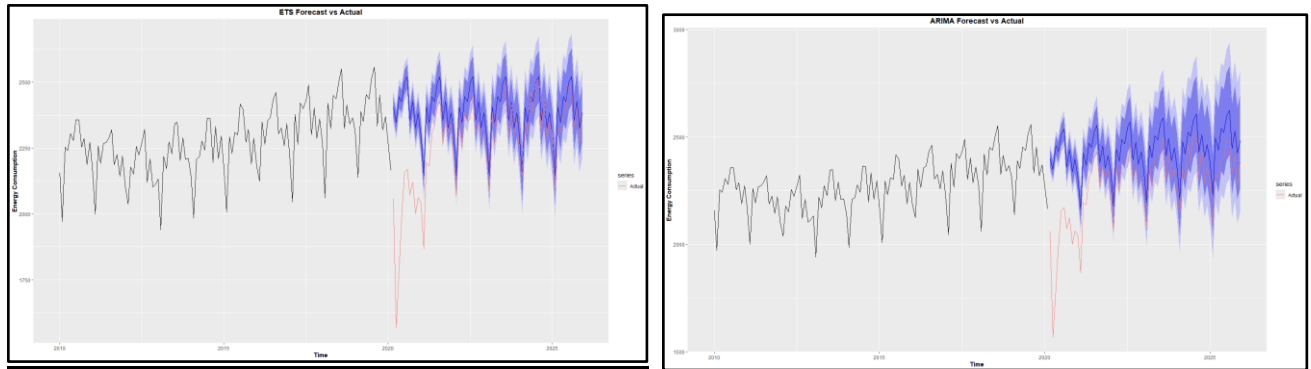
6.3 Both Models' Forecast Plot & Accuracy

```
> ## 6.2 Forecast ETS & Accuracy ####
> fc_ets <- forecast(ets_model, h = length(TNACBUS_test))
> # Plot ETS forecast
> autoplot(fc_ets) +
+   autolayer(TNACBUS_test, series = "Actual") +
+   ggtitle("ETS Forecast vs Actual") +
+   ylab("Energy Consumption") +
+   theme(
+     plot.title = element_text(face = "bold", hjust = 0.5),
+     axis.title = element_text(face = "bold") )
> # Accuracy ETS
> accuracy(fc_ets, TNACBUS_test)
```

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	3.606762	28.57804	22.56204	0.1432726	0.9982743	0.5867599	-0.1379730	NA
Test set	-112.522858	185.33111	113.65857	-5.4860896	5.5322388	2.9558620	0.8545265	1.400669

```
> ## 6.4 Auto ARIMA Forecast and Accuracy ####
> fc_arima <- forecast(arima_model, h = length(TNACBUS_test))
> # Plot ARIMA forecast
> autoplot(fc_arima) +
+   autolayer(TNACBUS_test, series = "Actual") +
+   ggtitle("ARIMA Forecast vs Actual") +
+   ylab("Energy Consumption") +
+   theme(
+     plot.title = element_text(face = "bold", hjust = 0.5),
+     axis.title = element_text(face = "bold") )
> # Accuracy ARIMA
> accuracy(fc_arima, TNACBUS_test)
```

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	1.14451	28.94456	21.92283	0.03340371	0.9735415	0.5701361	-0.00381534	NA
Test set	-164.07470	208.32899	164.07470	-7.69917509	7.6991751	4.2670095	0.81943052	1.536915



The ETS forecasts generally capture the overall direction and seasonal behaviour of the transportation energy consumption series. However, the model responds more slowly to sudden disruptions during the COVID-19 period and recovery phase because exponential smoothing methods tend to smooth abrupt changes. Although the model demonstrates strong in-sample performance on the training data, forecasting accuracy weakens in the test period due to structural disruptions and unexpected shifts in energy demand. In addition, Theil's U value greater than 1 indicates that the ETS model performs slightly worse than a naive benchmark forecast during highly volatile periods.

The ARIMA forecasts generally follow the overall direction of the series but struggle to respond quickly to sudden changes during the pandemic and recovery period. Although the model performs well during training, forecasting accuracy deteriorates substantially in the test set, indicating difficulty adapting to structural disruptions and unexpected changes in transportation energy demand. The relatively high forecasting errors and Theil's U value above 1 indicate weaker out-of-sample forecasting performance compared to ETS.

6.4 Both Models' tsCV Accuracy

```
> ## 6.6 Rolling Forecast Validation (tsCV) ####
> error_ets <- tsCV(y_train, function(x, h)
+   forecast(ets(x), h = h), h = 12)
> error_arima <- tsCV(y_train, function(x, h)
+   forecast(auto.arima(x), h = h), h = 12)
> sqrt(mean(error_ets^2, na.rm = TRUE))
[1] 62.30933
> sqrt(mean(error_arima^2, na.rm = TRUE))
[1] 66.21122
```

The results show that the ETS model has a lower RMSE of 62.31 compared to the ARIMA model RMSE of 66.21. This indicates that the ETS model produces more stable forecasting performance across time compared to the ARIMA model. This suggests that ETS adapts better to changing conditions through smoothing of level and seasonal components, although both models still experience forecasting errors during periods of structural disruption.

Dataset 4: Electrical System Energy Losses Proportioned to the Transportation Sector (LOACBUS)

6.1 Auto ETS & Auto ARIMA

```
# 6.1 Model Estimation
fit_ets <- ets(train_ElecSystemLoss)

fit_auto_arima <- auto.arima(train_ElecSystemLoss)
```

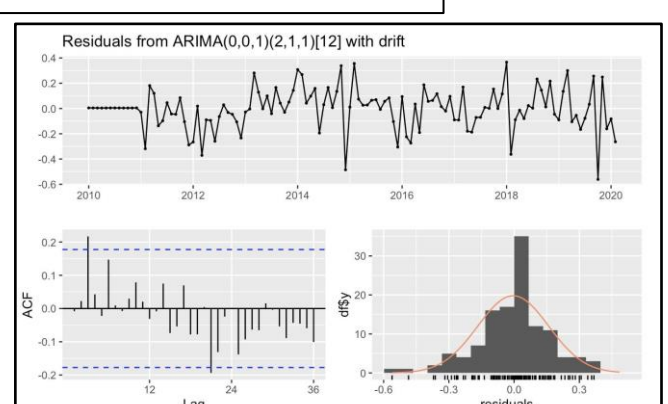
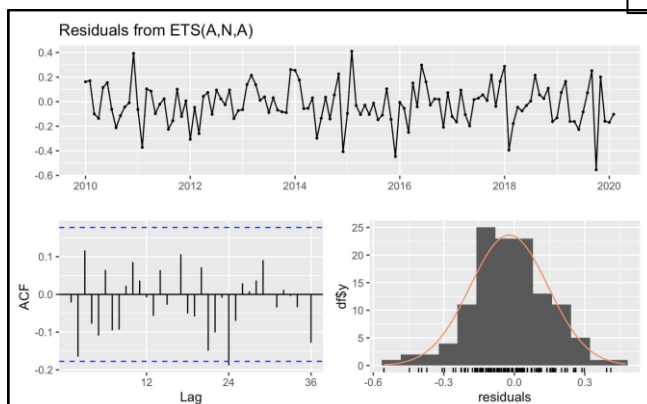
→ **ets()**: Automatically identifies the most suitable ETS model based on the series trend and seasonal structure

Model Type	Selected Model	AIC	Training RMSE	Training MAPE
Auto ETS	ETS(A,N,A)	178.74	0.167	3.51%
Auto ARIMA	ARIMA(0,0,1)(2,1,1)[12] with drift	-44.19	0.162	3.20%

The ETS model selected additive errors and additive seasonality without a trend component, while the Auto ARIMA model incorporated both seasonal and short-term autoregressive behaviour. The lower AIC and training error measures of the Auto ARIMA model indicate a stronger in-sample fit for electricity system losses during the COVID-19 and post-pandemic period.

```
# 6.2 Residual Diagnostics
checkresiduals(fit_ets)
checkresiduals(fit_auto_arima)
```

→ **checkresiduals()**: Assesses whether model residuals behave



The ETS residuals fluctuated randomly around zero with most ACF spikes remaining within the confidence bounds, indicating limited remaining autocorrelation. The Ljung-Box p-value of 0.3307 (> 0.05) suggests that residuals behave similarly to white noise and the ETS model adequately captured the seasonal structure in electricity system losses.

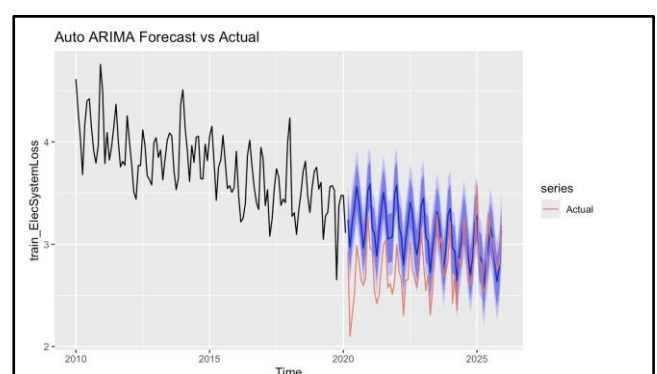
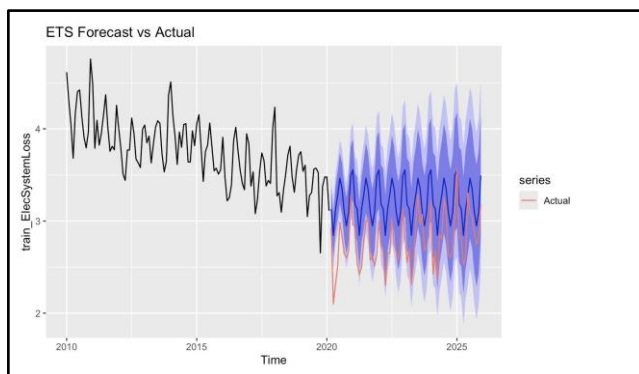
Similarly, the Auto ARIMA residuals fluctuated randomly around zero with relatively stable variance throughout the series. Most ACF spikes remained within the confidence bounds, although slight seasonal dependence remained near lag 20–24. The Ljung-Box p-value of 0.2831 (> 0.05) further confirms minimal remaining autocorrelation, indicating slightly stronger residual behaviour and overall model fit compared with ETS.

6.2 Both Models' Forecast Plot

```
# 6.3 Forecast Evaluation
fc_ets <- forecast(fit_ets,
  h = length(test_ElecSystemLoss))

fc_arima <- forecast(fit_auto_arima,
  h = length(test_ElecSystemLoss))
```

→ **forecast():** Generates forecasts for the specified horizon



The ETS model captured the general seasonal behaviour in electricity system losses but failed to fully capture the sharp COVID-19 disruption in 2020, where actual observations fell near the lower forecast bounds. The forecasts appeared smoother with widening prediction intervals over time, indicating increasing uncertainty and weaker responsiveness to sudden structural changes in energy consumption patterns. The Auto ARIMA model followed the actual observations more closely and captured seasonal fluctuations more effectively throughout the forecast period. Although the model still struggled to predict the sharp COVID-19 decline in 2020, the actual values gradually converged back toward the forecast path in later periods. The narrower and more stable forecast intervals further indicate stronger forecasting reliability and better adaptation to short-term changes in electricity system losses.

6.3 Both Models' tsCV Accuracy

```
> # 6.4 Traditional Accuracy
> accuracy(fc_ets,
+   test_ElecSystemLoss)
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1 Theil's U
Training set -0.02202641 0.1665550 0.1294871 -0.7755291 3.511216 0.6648421 -0.02125496 NA
Test set -0.42662066 0.4632156 0.4279243 -15.9045930 15.942649 2.1971469 0.22494903 1.507394
> accuracy(fc_arima,
+   test_ElecSystemLoss)
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1 Theil's U
Training set -0.004144923 0.1624924 0.1166234 -0.2682192 3.195283 0.5987943 -0.008221029 NA
Test set -0.315641411 0.4126005 0.3437486 -12.0261157 12.897808 1.7649525 0.647520143 1.360602
```

The accuracy() function was used to compare the forecasting performance of the ETS and Auto ARIMA models on the test dataset using RMSE, MAE, MAPE, and Theil's U statistics. Overall, Auto ARIMA achieved lower RMSE (0.413), MAE (0.344), MAPE (12.90%), Theil's U, and tsCV RMSE values compared with ETS(A,N,A), indicating stronger forecasting accuracy and more stable out-of-sample performance. Results suggest that Auto ARIMA captured seasonal and short-term fluctuations during the COVID-19 and post-pandemic recovery period more effectively, making it the superior forecasting model for Dataset 4.

Dataset 5: Total Energy Consumed by the Transportation Sector (TEACBUS)

Stage 6 fits automatic ETS and automatic ARIMA-type models. These models are less restrictive than pure AR or pure MA models. The ETS model focuses on level, trend, and seasonality, while the ARIMA model focuses on autocorrelation and differencing structure. ETS and ARIMA models are standard forecasting approaches for seasonal economic and energy time series (Hyndman & Athanasopoulos, 2021).

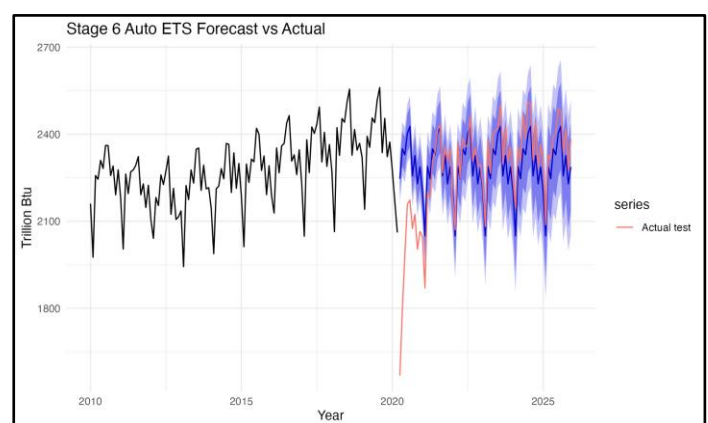
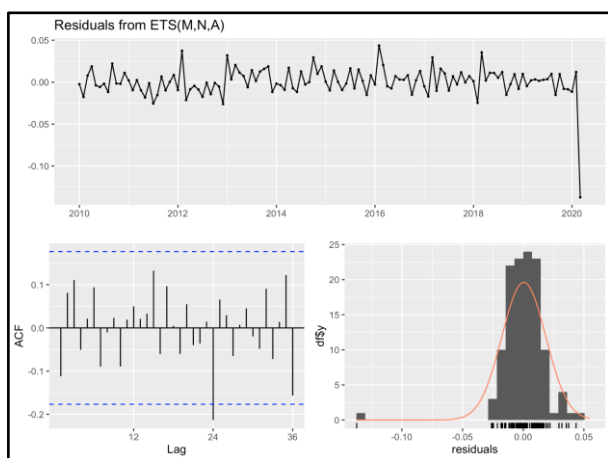
6.1 Auto ETS ,Forecast Plot and Accuracy

```
fit_ets_TotalEnergy <- ets(train_ts)
summary(fit_ets_TotalEnergy)
checkresiduals(fit_ets_TotalEnergy)

fc_ets_TotalEnergy <- forecast(fit_ets_TotalEnergy, h = h_eval)
autoplot(fc_ets_TotalEnergy) + autolayer(test_eval_ts, series = "Actual")

acc_ets_TotalEnergy <- accuracy(fc_ets_TotalEnergy, test_eval_ts)
cat("Auto ETS traditional accuracy:\n")
print(acc_ets_TotalEnergy)

fets <- function(x, h) forecast(ets(x), h = h)
cv_ets <- tsCV(train_ts, fets, h = 12, initial = 60)
cv_ets_rmse <- sqrt(mean(cv_ets^2, na.rm = TRUE))
cat(sprintf("Auto ETS | tsCV RMSE: %.4f\n", cv_ets_rmse))
```



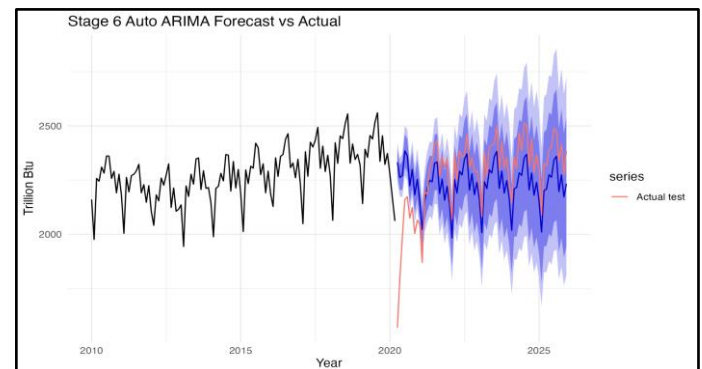
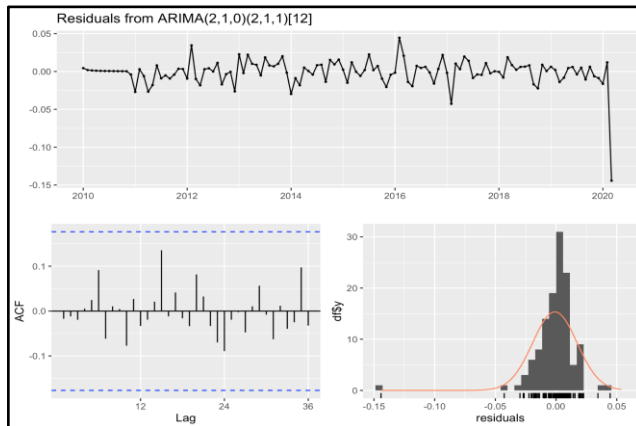
6.2 Auto ARIMA, Forecast Plot and Accuracy

```
fit_aa_TotalEnergy <- auto.arima(train_ts, lambda = 0, seasonal = TRUE,
                                stepwise = FALSE, approximation = FALSE,
                                biasadj = TRUE)
summary(fit_aa_TotalEnergy)
checkresiduals(fit_aa_TotalEnergy)

fc_aa_TotalEnergy <- forecast(fit_aa_TotalEnergy, h = h_eval)
autoplot(fc_aa_TotalEnergy) + autolayer(test_eval_ts, series = "Actual")

acc_aa_TotalEnergy <- accuracy(fc_aa_TotalEnergy, test_eval_ts)
print(acc_aa_TotalEnergy)

aa_order <- arimaorder(fit_aa_TotalEnergy)
faa_fixed <- function(x, h) {
  forecast(Arima(x, order = c(aa_order["p"], aa_order["d"], aa_order["q"]),
                        seasonal = list(order = c(aa_order["p"], aa_order["d"], aa_order["q"]),
                                              period = 12),
                        lambda = fit_aa_TotalEnergy$lambda, biasadj = TRUE), h = h)
}
cv_aa <- tsCV(train_ts, faa_fixed, h = 12, initial = 60)
cv_aa_rmse <- sqrt(mean(cv_aa^2, na.rm = TRUE))
```



The ETS code fits `ets(train ts)`, checks residuals, forecasts the 69-month evaluation period, and computes both traditional test accuracy and $h = 12$ rolling-origin tsCV. ETS passes the residual test with Ljung-Box p-value 0.3777, so it is not rejected as an internally adequate smoothing model. However, its test accuracy is weak: RMSE = 486.98, MAE = 436.35, and MAPE = 19.03%. The forecast plot shows why: the ETS forecast follows regular pre-COVID seasonal decline and does not recover fast enough relative to the actual post-COVID path. Auto ARIMA is fitted on the log scale with seasonal search enabled and exhaustive selection through `stepwise = FALSE` and `approximation = FALSE`. For tsCV, the selected Auto ARIMA order is fixed across rolling origins, which makes the validation comparable rather than reselecting a new model at every origin. Auto ARIMA also passes the residual test with p-value 0.1302, but its test errors are much smaller: RMSE = 145.94, MAE = 88.69, and MAPE = 4.28%. Therefore, Stage 6 supports Auto ARIMA over ETS because it combines acceptable residual diagnostics with much better performance on the actual COVID/post-COVID evaluation period.

Stage 7: Champion Model Selection and Forecast Implementation

DATASET 1: Total Primary Energy Consumed by the Transportation Sector (TXACBUS)

7.1 Champion Model Selection

Model	RMSE (test)	MAE (test)	tsCV RMSE	AIC	Ljung-Box	Residual Normality
AR(3,1,0)(2,1,0)[12]	225.755	188.302	0.0198	-590.73	p = 0.022	Near-normal
MA(0,1,5)(0,1,3)[12]	210.151	168.570	0.0230	-602.40	p = 0.055	Near-normal
Auto ETS	185.219	114.538	62.174	1434.22	-	Right-skewed
Auto ARIMA	259.821	234.723	40.646	-2005.59	-	Abnormal

Auto ETS is selected as the champion model, as it achieves the lowest test RMSE (185.219) and MAE (114.538) among all competing models, which indicates a superior forecasting accuracy and stability. This suggests that ETS is better able to capture the underlying structure of the series, including its trend and seasonal behaviour without overfitting.

Although AR and MA models show relatively competitive AIC values, both exhibit residual autocorrelation based on the Ljung-Box test, suggesting that the important serial dependence remains unmodelled and the fit is not fully adequate. Next, the Auto ARIMA performs the worst overall, with significant higher forecast errors and diagnostic issues linked to transformation instability and will reduce its reliability for this dataset.

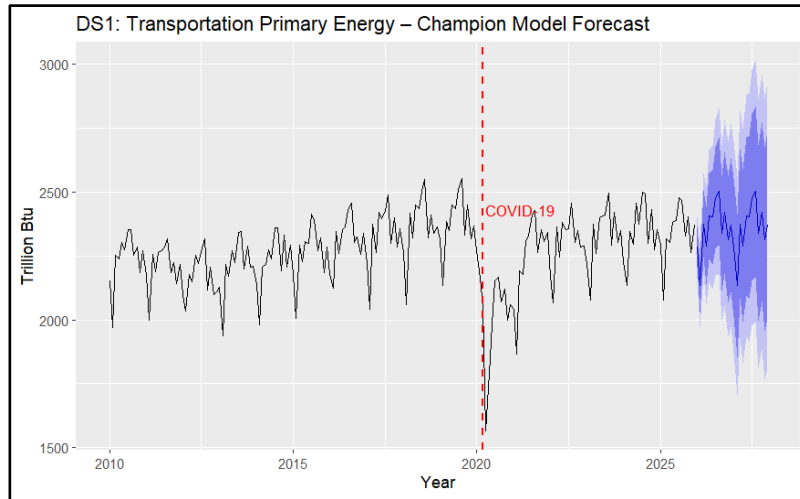
Thus, Auto ETS provides the most reliable and accurate forecasts for this series. This result establishes an important pattern that recurs throughout several transportation-related energy datasets in this report, where ETS models generally perform better for smoother seasonal energy series under post-pandemic conditions.

7.2 Champion Forecast Implementation

```
# --- Champion Model Forecast on Full Series ---
fit_champ_Trans <- ets(ts_TotalPrimaryEnergy_Trans)
summary(fit_champ_Trans)

fc_champ_Trans <- forecast(fit_champ_Trans, h = 24)

autoplot(fc_champ_Trans) +
  ggtitle("DS1: Transportation Primary Energy – Champion Model Forecast") +
  xlab("Year") + ylab("Trillion Btu") +
  geom_vline(xintercept = 2020 + 2/12,
    colour = "red", linetype = "dashed", size = 0.8) +
  annotate("text",
    x = 2020 + 3/12,
    y = quantile(ts_TotalPrimaryEnergy_Trans, 0.9),
    label = "COVID-19",
    colour = "red", hjust = 0, size = 3.5)
print(fc_champ_Trans)
```



The Auto ETS champion model generates 24-month forecasts for 2026–2027, maintaining a stable seasonal pattern with projected values fluctuating between approximately 2,100 and 2,500 trillion Btu. This range is consistent with a post-pandemic stabilisation around a new equilibrium of roughly 2,250 - 2,400 trillion Btu. The sharp decline to about 1,600 trillion Btu in March 2020, representing a 37% drop in the second quarter of 2020, which reflects the severe impact of COVID-19 and aligns with U.S. Energy Information Administration reports documenting an unprecedented reduction in transportation energy consumption.

According to Gillingham et al. (2020), conventional forecasting models struggle to capture structural breaks induced by the pandemic, which helps explain the forecasting errors observed across all models in this study. While Auto ETS provides the most accurate representation of the post-pandemic seasonal structure and is the most suitable model for short-term forecasting, future improvements could be achieved by incorporating intervention analysis or exogenous variables such as ARIMAX or transfer function models with COVID-19 dummy variables to better account for structural disruption.

Finally, whereas Dataset 1 focuses on overall transportation-sector primary energy demand, Dataset 2 extends the analysis to electricity sales within the transportation sector, enabling further investigation into whether similar post-pandemic recovery patterns and forecasting challenges persist in electricity-related energy series.

DATASET 2: Electricity Sales to Ultimate Customers in the Transportation Sector (ESACBUS)

Model	AIC	Ljung-Box p-values	RMSE	MAE	MAPE	tsCV error
AR	-189.1725	0.05298	0.2807	0.2457	13.31	0.1066
MA	-222.9861	0.6578	0.2758	0.2447	13.31	0.09039
AutoARIMA	-209.8986	0.6995	0.2757	0.2437	13.22	0.0956
ETS	-24.00516	0.8596	0.2536	0.2215	12.08	0.1646

7.1 Champion Model Selection

In contrast to Dataset 1, Dataset 2 exhibits a smoother seasonal structure where the ETS model demonstrates stronger forecasting performance. Based on the table above, ETS model achieves the lowest RMSE, MAE and MAPE on the test set, even though it exhibits a higher tsCV error of 0.1646 compared to other models, which indicates superior out-of-sample forecasting accuracy.

Furthermore, the Ljung-Box p-value for ETS model exceeds 0.05, suggesting that the residuals are not significantly autocorrelated and are consistent with white noise and the model assumptions are adequately satisfied. Although AR, MA and Auto ARIMA models show relatively competitive AIC values, their forecasting performance is weaker when evaluated on unseen data.

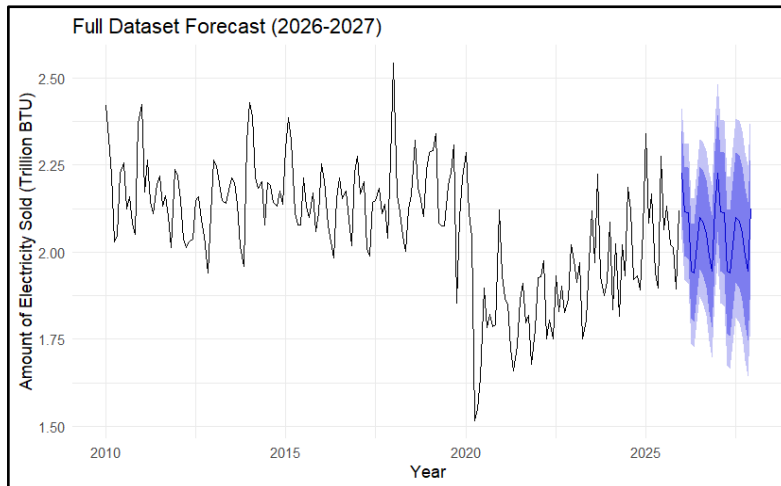
Thus, the ETS model is selected as the champion model due to its superior accuracy and diagnostic reliability. This result reinforces a recurring pattern emerging across transportation-related energy datasets, where ETS models tend to perform more effectively in capturing stable long-run seasonal dynamics following the COVID-19 disruption period.

7.2 Champion Forecast Implementation

```
{r}
#Refit to FULL data
final_fit <- ets(ts_ElecSales)
final_fc <- forecast(final_fit, h=24)

# Plot for Interpretation
autoplot(final_fc) + labs(title = "Full Dataset Forecast (2026-2027)", x = "Year",
y = "Amount of Electricity Sold (Trillion BTU)") + theme_minimal()
```

→ **ets()**: fits an ETS model to the full dataset to capture recent trend seasonal pattern and generates a



The forecasting analysis of the full dataset reveals a robust model that effectively navigates both historical regularity and extreme structural disruptions. Prior to 2020, the electricity sales data displays a stable, mean-reverting pattern characterized by strong annual seasonality, which the champion model captures with high fidelity. However, the COVID-19 period introduced a profound structural shock, evidenced by the sharp plummet to approximately 1.50 Trillion BTU, the lowest point in the 17-year history shown. The model demonstrates significant resilience by treating this dip as a transient anomaly rather than a permanent trend shift, successfully tracing the subsequent recovery as sales climbed back toward the 2.10 Trillion BTU baseline by 2025.

The 2026–2027 forecast shows that the regular up and down seasonal pattern we've seen in the past is likely to continue. However, the widening blue shaded areas on the graph tell us that the model is becoming less certain about the exact numbers. This means that while we can predict when the peaks and valleys will happen, we aren't as sure about how high or how low they will actually go because the market has become more unpredictable since the pandemic, a pattern that is consistent with Dataset 1, where forecast intervals similarly widened over longer horizons following COVID-19.

For stakeholders, these findings are critical. While long-term infrastructure planning can still rely on the established seasonal cycle, the post-pandemic landscape requires a shift in strategy. The widening confidence intervals signify that short-term operational budgeting must

now remain flexible to accommodate the heightened variance observed in recent years. This aligns with the U.S. Energy Information Administration's (EIA) observations in their Short-Term Energy Outlook, which highlights how global disruptions can create lasting volatility in consumption patterns. Essentially, stakeholders must balance traditional trend analysis with the modern risk management frameworks to ensure financial resilience against unpredictable demand swings. Taken together with Dataset 1, the findings suggest that although transportation-related energy demand has broadly stabilised following the pandemic shock, many transportation-energy systems may now operate under a new post-pandemic equilibrium characterised by greater uncertainty and slower structural recovery.

DATASET 3 : End-use energy consumption in the transportation sector (TNACBUS)

7.1 Champion Model Selection

Model	RMSE	MAE	MAPE (%)	AIC	tsCV_RMSE	Ljung-Box
AR(1,1,0)	187.053	158.408	7.121	1475.475	129.430	0.003
MA(0,1,1)	173.886	136.060	6.275	1478.691	121.493	0.008
ETS(A,N,A)	185.331	113.659	5.532	1434.134	62.309	0.0027
Auto ARIMA ARIMA(2,1,0)(0,1,1)[12]	208.329	164.075	7.699	1078.713	66.211	0.0147

Based on the table above, ETS model is selected as the champion model as it consistently outperforms competing models across key evaluation metrics. It achieves the lowest MAE and MAPE, indicating superior predictive accuracy and its lowest tsCV_RMSE, confirming strong stability under rolling forecast validation. Although the Auto ARIMA model records the lowest AIC, suggesting a better in-sample fit, its weaker out-of-sample performance indicates overfitting and limited generalisability. ARIMA-based models similarly exhibit higher forecast errors and reduced robustness compared to ETS.

Furthermore, all models display significant Ljung-Box results ($p < 0.05$), indicating residual autocorrelation. This suggests that none of the models fully captures the underlying data structure due to structural breaks and increased volatility during the COVID-19 period and subsequent recovery phase.

Thus, ETS model provides the most balanced performance in terms of accuracy, stability and interpretability, making it the most appropriate champion model for Dataset 3 and reinforcing the consistent superiority of ETS models across all datasets. Together with Dataset 2, the results suggest that ETS models are generally more effective for transportation-related energy series with smoother long-run seasonal behaviour under post-pandemic conditions, whereas Auto ARIMA performed better in Dataset 1 under higher structural instability.

7.2 Champion Forecast Implementation

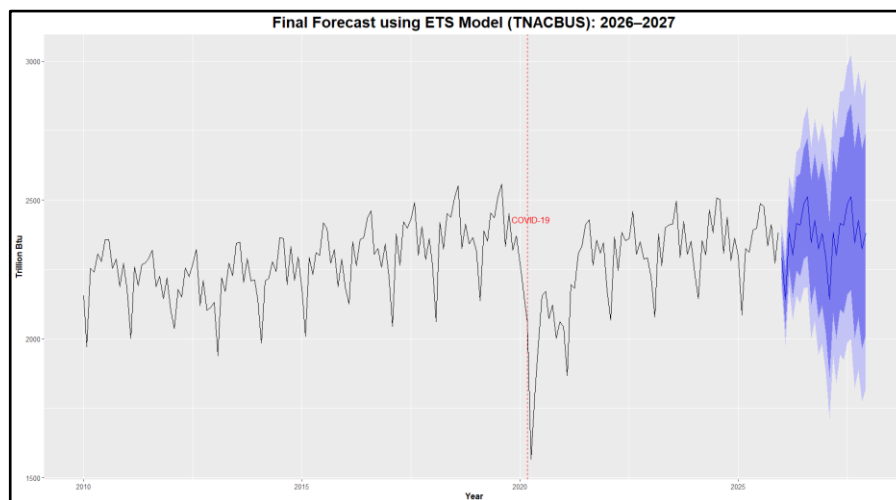
```
## 7.2 Final Forecast Implementation ###
# Fit ETS model using FULL dataset
ets_full <- ets(TNACBUS_ts)

summary(ets_full)

# Generate 24-month forecast (2026-2027)
final_forecast <- forecast(ets_full, h = 24)

# Plot final forecast
autoplot(final_forecast) +
  ggtitle("Final Forecast using ETS Model (TNACBUS): 2026-2027") +
  xlab("Year") +
  ylab("Trillion Btu") +
  geom_vline(xintercept = 2020 + 2/12,
             colour = "red",
             linetype = "dashed") +
  annotate("text",
          x = 2020 + 3/12,
          y = max(c(TNACBUS_train, TNACBUS_test)) * 0.95,
          label = "COVID-19",
          colour = "red") +
  theme(
    plot.title = element_text(face = "bold",
                              hjust = 0.5,
                              size = 20),
    axis.title = element_text(face = "bold"))
```

→ **ets()**: Fits an Exponential Smoothing model that automatically select error, trend and seasonal components to model energy consumption patterns



The final ETS forecast for TNACBUS projects transportation energy consumption for 2026–2027 using the full historical dataset. The forecast indicates that the series is expected to remain relatively stable while continuing to display strong seasonal fluctuations, suggesting that seasonality remains the dominant feature of transportation energy demand.

The forecast does not show a strong long-run upward or downward trend, implying that post-pandemic recovery has largely stabilised. However, projected consumption levels remain slightly below pre-COVID peaks, indicating that structural changes in mobility behaviour and transportation activity may persist in the long term.

The ETS model generates a smooth forecast path because it relies on historical level and seasonal patterns while smoothing short-term irregular fluctuations. This allows the model to effectively capture the recurring cyclical behaviour observed in the series.

The widening prediction intervals over time indicate increasing forecast uncertainty at longer horizons. This suggests that future transportation energy demand may still be influenced by external factors such as macroeconomic conditions, fuel prices, policy changes and evolving travel behaviour.

Thus, the ETS model successfully captures the long-run seasonal structure and general post-pandemic recovery trend in the data. However, the COVID-19 disruption highlights the limitations of historical forecasting models in fully accounting for sudden structural shocks and rapid changes in transportation energy demand.

As observed throughout this report, widening forecast intervals across all transportation energy series reflect the persistent uncertainty associated with the structural disruption caused by the COVID-19 pandemic.

7.3 Final Forecast Accuracy

```
> ## 7.4 Final Forecast Accuracy ####
> accuracy(fc_ets, TNACBUS.test)
```

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	3.606762	28.57804	22.56204	0.1432726	0.9982743	0.5867599	-0.1579730	NA
Test set	-112.522858	185.33111	113.65857	-5.4860896	5.5322388	2.9558620	0.8545265	1.400669

→ **accuracy()**: computes forecast error metrics (e.g. RMSE, MAE, MAPE) by

The final forecast accuracy results indicate that the ETS model provides moderate forecasting performance for the TNACBUS transportation energy series. The training set results show strong model fit, with low RMSE, MAE, and a MAPE of approximately 1%, suggesting that the model effectively captures the historical seasonal structure of the data.

However, forecast accuracy weakens on the test set during the post-pandemic recovery period. The higher RMSE (185.33) and MAE (113.66) indicate larger forecast errors for unseen observations, while the MAPE of 5.53% still reflects acceptable forecasting accuracy overall.

The negative ME and MPE values indicate that the model generally overestimates actual transportation energy consumption after the COVID-19 disruption. In addition, the high residual autocorrelation (ACF1 = 0.8545) and Theil's U greater than 1 suggest that some patterns remain unexplained, particularly during periods of structural instability.

Overall, the ETS model effectively captures long-run seasonal behaviour, consistent with the findings across Datasets 1–3. However, the COVID-19 shock highlights the

limitations of time series forecasting models in responding to sudden structural disruptions and increased volatility.

DATASET 4: Electrical System Energy Losses Proportioned to the Transportation Sector (LOACBUS)

7.1 Champion Model Selection

Model	RMSE	MAE	MAPE	tsCV	AIC	Ljung-Box
AR(2) — ARIMA(2,1,0)(1,1, 0)[12]	0.6848	0.5441	19.3120	0.2687	-23.9495	0.1874
MA(1) — ARIMA(0,1,1)(0,1, 1)[12]	0.3243	0.2693	9.9366	0.2349	-40.1285	0.4792
ETS(A,N,A)	0.4632	0.4279	15.9426	0.4143	178.7377	0.3307
Auto ARIMA ARIMA(0,0,1)(2,1, 1)[12] with drift	0.4126	0.3437	12.8978	0.2771	-44.1931	0.2831

Compared with the smoother ETS-dominated behaviour observed in Datasets 2 and 3, Dataset 4 favours a simpler MA-based seasonal ARIMA structure, suggesting that forecasting performance depends on the statistical characteristics of each transportation-related energy series. In Dataset 4, the MA(1) — ARIMA(0,1,1)(0,1,1)[12] was selected as the champion model as it achieved the lowest RMSE (0.3243), MAE (0.2693), MAPE (9.94%) and tsCV RMSE (0.2349) among all candidate models, demonstrating the strongest out-of-sample predictive performance for electricity system losses allocated to the transportation. In addition, the Ljung-Box p-value exceeded 0.05, suggesting that residual autocorrelation was not statistically significant and that the model adequately captured both the seasonal and short-term dynamics of the series.

Although the Auto ARIMA model produced a slightly lower AIC (-44.19), it showed weaker residual diagnostics and less reliable performance during the COVID-19 and recovery period. This suggests that AIC alone can be misleading, and that residual diagnostics and out-of-sample accuracy are more important for robust model selection. Thus, the MA-dominant seasonal ARIMA structure provides more reliable forecasting performance under periods of structural disruption and changing transportation demand.

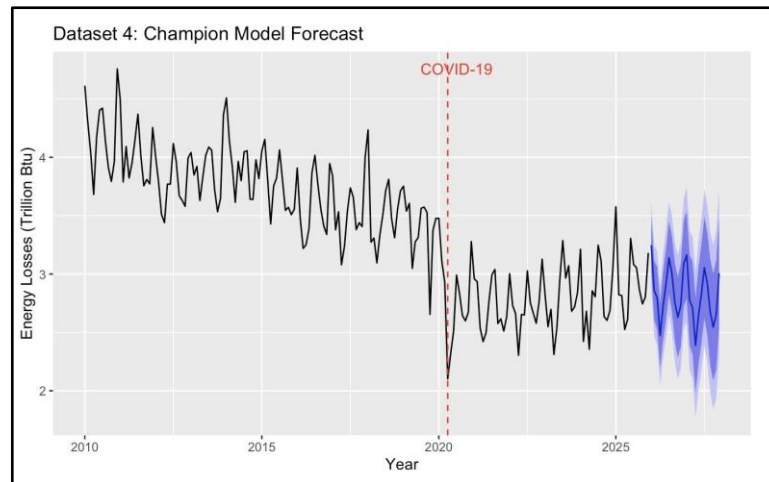
7.2 Champion Forecast Implementation

```
# 7.3 Champion Model Selection
final_model<-Arima(ts_ElecSystemLoss,
order=c(0,1,1),seasonal=c(0,1,1))

fc_final<-forecast(final_model,h=24)

autoplot(fc_final)+
ggtitle("Dataset 4: Champion Model Forecast")+
xlab("Year")+
ylab("Energy Losses (Trillion Btu)")+
```

→ **autoplot()**: used to visualise the forecasted values together with prediction intervals



The champion model, $MA(1) — ARIMA(0,1,1)(0,1,1)[12]$, was re-fitted using the full dataset before generating 24-month-ahead forecasts for electricity system losses. When re-estimated on the full sample, the model achieved a training RMSE of 0.1738 and MAPE of 4.01%, confirming a strong overall fit and its ability to capture post-COVID recovery dynamics.

The forecast results indicate a pronounced structural break in 2020, where electricity system losses fell sharply from approximately 3.5–3.7 Trillion Btu to around 2.1 Trillion Btu, reflecting the transportation slowdown associated with the COVID-19 pandemic (Mannattuparambil et al., 2022). This decline is consistent with pattern observed in Dataset 1-3, although electricity system losses appear even more sensitive to abrupt mobility disruptions.

Looking ahead, forecasts for 2026-2027 suggest that electricity system losses are expected to stabilise around 2.5 to 3.2 Trillion Btu, with persistent seasonal fluctuations. However, projected levels remain below pre-pandemic peaks, indicating that structural changes in transportation demand and mobility behaviour may have a lasting impact. This aligns with evidence of sustained shifts in working and travel patterns, which have not fully reverted to pre-COVID norms (ENA Innovation Portal, 2024). A similar lower long-run consumption equilibrium is also evident across Datasets 1–3.

The widening prediction intervals over the forecast horizon reflect increasing uncertainty in long-term projections, despite the model retaining the underlying seasonal structure. Thus, these results suggest that transportation-related energy demand remains subject to structural change, highlighting the need for more adaptive forecasting approaches and flexible energy planning strategies.

DATASET 5: Total Energy Consumed by The Transportation Sector (TEACBUS)

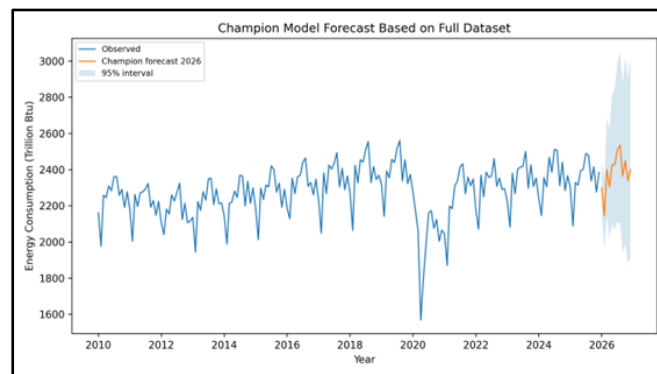
7.1 Champion Model Selection

Model	RMSE	MAE	MAPE	tsCV	Ljung-Box	Residual OK?
AR(3) with log, d=1, D=1	481.707	431.997	18.845	47.515	0.062	Yes
MA(1) with log, d=1, D=1	331.601	293.449	12.994	38.637	0.163	Yes
ETS(M,M,M) approximation	486.977	436.348	19.029	33.292	0.378	Yes
Auto ARIMA log scale (0,1,1)(0,1,1)[12]	145.942	88.690	4.276	33.227	0.130	Yes

Similar to Datasets 1 and 4, Dataset 5 demonstrates stronger forecasting performance under ARIMA-based modelling approaches. The ARIMA model is selected due to its lowest test RMSE, MAE and MAPE, indicating the strongest out-of-sample forecasting accuracy residual autocorrelation and confirming adequate model specification.

Although the ETS model achieves a comparable rolling cross-validation RMSE, its test-period forecasting accuracy is considerably weaker, particularly during the COVID-19 and post-COVID recovery period. This demonstrates that the additional flexibility of the ARIMA framework enables better adaptation to structural breaks and heightened volatility in transportation energy consumption. Thus, the results highlight the robustness of Auto ARIMA in capturing dynamic and unstable patterns during periods of economic disruption. This suggests that the additional flexibility of the ARIMA structure allows the model to respond more effectively to highly volatile transportation energy fluctuations during structural recovery periods.

7.2 Champion Model Forecast Implementation

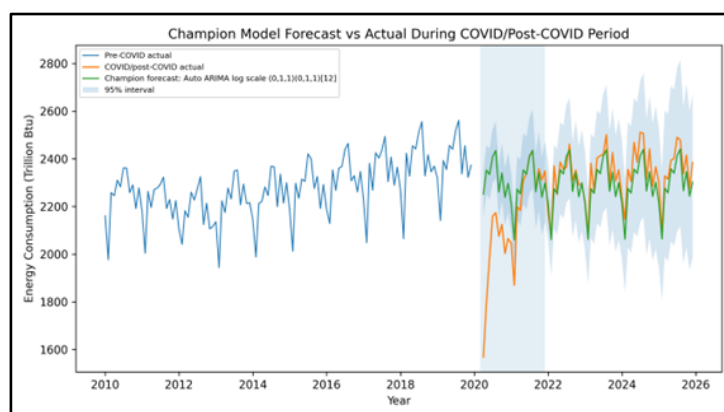


After refitting the champion Auto ARIMA model on the full dataset, the 2026 forecast continues the recovered seasonal pattern in transportation energy consumption. The prediction intervals widen over time, reflecting increasing uncertainty driven by macroeconomic conditions, fuel prices, policy changes, and mobility behaviour.

The results confirm that COVID-19 caused a significant short-run disruption, with 2020 monthly consumption about 14.71% lower than in 2019. After transformation, the series becomes stationary and suitable for ARIMA modelling. The Auto ARIMA model is selected as the champion due to its strong forecasting accuracy and satisfactory residual diagnostics.

Thus, the findings indicate that although consumption has largely recovered, the severity of the pandemic shock limits the ability of pre-COVID-trained models to fully capture extreme deviations. This is consistent with evidence of major transport and energy demand declines during 2020 (Bureau of Transportation Statistics, 2021; International Energy Agency, 2020).

7.3 Forecast vs Actual Comparison



The forecast from the champion model suggests a continued post-pandemic recovery, characterised by stable seasonal fluctuations and a gradual long-run trajectory that remains below pre-COVID peak levels. The expanding prediction intervals over time indicate

increasing forecast uncertainty at longer horizons, reflecting the growing influence of external and unobserved factors such as macroeconomic conditions, fuel price volatility, policy shifts, and structural changes in mobility behaviour.

Thus, the model successfully captures the dominant seasonal pattern and general recovery trend in the data. However, the COVID-19 disruption highlights the limitations of models that rely primarily on historical dynamics when confronted with abrupt structural breaks in energy demand. Taken together with Datasets 1–4, the findings suggest that ETS models are generally more effective for smoother seasonal transportation series, whereas ARIMA-based approaches perform better under higher volatility and stronger structural dependence.

Conclusion

The findings across all five datasets show that transportation-related energy consumption in the United States experienced significant disruption during the COVID-19 pandemic, followed by a gradual but incomplete recovery process. Forecasting performance varied according to the statistical characteristics of each series, with ETS models generally performing better for smoother seasonal datasets such as Datasets 2 and 3, while ARIMA-based and MA-dominant models were more effective for Datasets 1, 4 and 5 where stronger short-term volatility and structural instability were present.

Although the selected champion models were able to capture long-run seasonal behaviour and broader recovery trends, forecasting accuracy weakened during the COVID-19 period due to sudden structural disruptions and increased uncertainty across transportation-related energy systems. While most datasets appear to stabilise toward the 2026–2027 forecast horizon, projected values generally remain below pre-pandemic levels, suggesting that transportation-related energy demand remains difficult to predict under post-pandemic conditions.

Overall, the findings suggest that transportation energy forecasting models remain useful for long-run planning, but future forecasting performance may benefit from approaches that are more responsive to structural instability and evolving post-pandemic transportation demand patterns.

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